



AN INTELLIGENT FRAMEWORK FOR EV BATTERY HEALTH MONITORING AND LIFECYCLE MANAGEMENT

Submitted In Partial Fulfilment of Requirements
For the Degree Of
Bachelor of Technology
(Artificial Intelligence and Data Science)
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ABSTRACT

Electric Vehicle (EV) batteries are the most critical and expensive component in modern electric mobility, yet no standardized, vendor-agnostic framework exists to monitor battery health, degradation, and lifecycle performance across diverse fleet configurations. Although large volumes of battery data are generated by Battery Management Systems (BMS), CAN bus interfaces, and charging infrastructure, this data remains fragmented and underutilized, limiting actionable insights for OEMs, fleet operators, and end users. This project presents an Intelligent Framework for EV Battery Health Monitoring and Lifecycle Management, developed in collaboration with Magenta EV Solutions Pvt. Ltd., which bridges the gap between laboratory-grade battery research and real-world deployment conditions.

The framework is organized into four interconnected analytical modules. The first module addresses Remaining Useful Life (RUL) prediction and knee-point detection, employing classical machine learning models (Ridge Regression, Random Forest, XGBoost) and deep learning architectures (LSTM, GRU, CNN-LSTM) on structured charge cycle data. A piecewise linear MSE minimization approach is used to identify the knee point — the cycle at which irreversible accelerated capacity fade begins. Physical consistency is enforced via monotonicity constraints and SEI-layer noise removal. XGBoost achieved the best single-model performance with test $R^2 = 0.83$ and RMSE = 11.2 cycles, while the weighted LSTM-GRU-XGBoost ensemble delivered robust generalization across unseen battery IDs.

The second module implements a Behavior-Aware Battery Management System (BA-BMS), which fuses real-time driver telemetry with electrochemical state estimation. Two composite indices are engineered: the Aggressiveness Index (AI), which quantifies driver behavior on a normalized 0-to-1 scale, and the Battery Stress Index (BSI), which captures electrochemical stress from current spikes, temperature variation, and voltage fluctuations. LightGBM achieved the best performance among all evaluated models with $R^2 = 0.2548$, RMSE = 9.10, and MAE = 6.11.

The third module focuses on real-world State of Health (SOH) estimation and Thermal Risk Classification using large-scale fleet telemetry exceeding 50 million records. A Hybrid CNN-LSTM architecture achieved the best SOH estimation with RMSE = 5.29%,

while Random Forest delivered flawless thermal safety classification with F1-Score = 0.997.

The fourth module provides unified EV battery analytics covering State of Charge (SOC), SOH, RUL, and Mileage per Charge estimation without data leakage. Tree-based ensemble models dominated all tasks: Random Forest achieved $R^2 = 0.9973$ for SOC, Extra Trees achieved $R^2 = 0.9990$ for SOH, Random Forest achieved $R^2 = 1.0000$ for RUL, and XGBoost achieved $R^2 = 0.9466$ for Mileage estimation.

The framework demonstrates that integrating physics-informed feature engineering, domain-expert behavioral indices, and ensemble machine learning delivers a scalable, interpretable, and practically deployable battery intelligence platform applicable across multiple EV classes, battery chemistries, and usage profiles.

Key words: Battery Management System, State of Health, Remaining Useful Life, Knee-Point Detection, Driving Behaviour Analysis, Electric Vehicle, XGBoost, LSTM, CNN-LSTM, Random Forest, LightGBM, Lifecycle Management.

Contents

List of Figures	viii
List of Tables.....	ix
Nomenclature	x
1 Introduction	1
1.1 Problem Definition.....	1
1.2 Motivation.....	4
1.3 Scope.....	4
1.4 Salient Contributions	5
1.5 Organization of the Report.....	5
2 Literature Survey	6
3 Software Project Management Plan	9
3.1 Introduction.....	9
3.1.1 Project Overview	10
3.1.2 Project Deliverables	11
3.2 Project Organization	10
3.2.1 Software Process Model.....	10
3.2.2 Roles and Responsibilities.....	11
3.2.3 Tools and Techniques.....	11
3.3 Project Management Plan	11
3.3.1 Tasks.....	11
3.3.1.1 Description.....	11
3.3.1.2 Deliverables and Milestones	13
3.3.1.3 Resources Needed.....	13
3.3.1.4 Dependencies and Constraints	14
3.3.1.5 Risks and Contingencies	14
3.3.2 Time Table (Project Schedule – Semester VI)	15
4 Software Requirements Specification	16
4.1 Introduction.....	16
4.1.1 Product Overview	16
4.2 Specific Requirements	16
4.2.1 External Interface Requirements	16
4.2.2 Functional Requirements.....	17
4.2.3 Non-Functional Requirements.....	18
5 System Design.....	20
5.1 Technical Workflow and System Architecture	20
5.2 Dataset Description.....	21
5.2.1 Charge Cycles Dataset.....	21
5.2.2 Features RUL Dataset	21
5.2.3 Device Telemetry Dataset	21

5.2.4 Trip Logs Dataset	22
6 Module I: Knee-Point Detection and RUL Prediction	23
7 Module II: Driving Behaviour Analysis (BA-BMS).....	34
8 Module III: Battery Health (SOH) Estimation and Thermal Management	43
9 Module IV: EV Battery Analysis – SOC, SOH, RUL and Mileage Estimation	49
10 Conclusions and Future Work.....	56
References	57
Nomenclature	63

LIST OF FIGURES

3.3	Project Schedule Gantt Chart	15
5.1	Technical Workflow	20
6.1	Data Preprocessing Module I	23
6.2	Battery Degradation Curve.....	27
6.3	Piecewise Degradation Analysis	27
6.4	Knee Detection Cell RZB03039.....	28
6.5	Knee Detection Cell RZD10213.....	28
6.6	Model Architecture Module I.....	29
6.7	RUL Performance Metrics.....	30
6.8	RUL Prediction Performance	30
6.9	XGBoost Feature Importance.....	32
7.1	Data Preprocessing Module II	35
7.2	Driver Behaviour Analysis	37
7.3	Correlation Heatmap.....	37
7.4	SOH Distribution.....	38
7.5	SHAP Values LightGBM	38
7.6	Model Architecture Module II.....	39
7.7	Driving Behaviour Model Comparison	40
8.1	Data Preprocessing Module III.....	44
8.2	SOH Model Performance	46
8.3	Thermal Management Results.....	48
9.1	Model Architecture Module IV	50
9.2	SOC Prediction Performance.....	51
9.3	SOH Prediction Performance	52
9.4	RUL Prediction Performance	53
9.5	Mileage Prediction Performance	54

LIST OF TABLES

6.1	RUL Model Performance	30
6.2	XGBoost Feature Importance	31
7.1	Driving Behaviour Model Performance	39
7.2	LightGBM Feature Importance	40
8.1	SOH Model Comparison	45
8.2	Thermal Management Model Comparison.....	47
9.1	SOC Model Performance.....	50
9.2	SOH Model Performance	51
9.3	RUL Model Performance	52
9.4	Mileage Model Performance	53

CHAPTER 1 — INTRODUCTION

1.1 Problem Definition

The global transition toward electric mobility has positioned the lithium-ion battery pack as the single most critical and cost-intensive component in the electric vehicle ecosystem. A battery pack in a modern electric vehicle account for a substantial fraction of the total vehicle cost and determines the usable range, operational reliability, and overall ownership experience. Despite this centrality, the current state of battery monitoring and lifecycle management is deeply fragmented, technically opaque, and largely incapable of delivering the kind of transparent, actionable intelligence that fleet operators, original equipment manufacturers, insurance companies, warranty departments, and end consumers actually require.

At the heart of this problem lies a fundamental disconnect between data availability and data utility. Modern electric vehicles are instrumented with Battery Management Systems that continuously monitor cell-level voltage, pack-level current, temperature distributions, and State of Charge. CAN bus interfaces transmit this telemetry in real time to onboard controllers. Charging systems log energy throughput, session durations, and thermal profiles. The result is a continuous, high-resolution stream of operational data generated across every trip, every charge cycle, and every parking event across the entire vehicle lifetime. Despite this richness, the data remains siloed within proprietary OEM systems, inconsistently formatted across vendors, and largely unanalysed beyond simple threshold-based alerting. No standardized framework exists to translate this raw operational data into reliable, vendor-agnostic estimates of battery State of Health, Remaining Useful Life, degradation velocity, or thermal risk.

This creates a cascade of practical problems that affect every stakeholder in the electric vehicle value chain. Fleet operators managing large numbers of electric vehicles face unplanned downtime caused by battery failures they could not anticipate, because their systems lacked the predictive intelligence to flag accelerating degradation before it became a service event. They also face suboptimal charging strategies because no system is integrating driving behavior data with battery state to recommend charging windows that would maximize cycle life. Warranty departments at OEMs face disputes over battery degradation claims they cannot resolve objectively, because there is no auditable, physics-

grounded record of how a specific battery aged under specific usage conditions. End consumers have no visibility into the real-world State of Health of the battery in a vehicle they are considering purchasing, creating asymmetric information that suppresses the secondary market for electric vehicles. Insurers have no quantitative basis for pricing battery replacement risk into electric vehicle policies.

The technical root of all these problems is the same: the absence of a unified, data-driven, physically consistent, vendor-agnostic battery intelligence platform that can ingest heterogeneous operational data, apply rigorous machine learning and deep learning methods, and produce reliable, interpretable estimates of battery condition across the full lifecycle.

This project directly addresses this gap. The framework developed here was formulated in collaboration with Magenta EV Solutions Pvt. Ltd., a company operating real-world electric vehicle fleets and managing battery assets at scale. The problem definition was therefore not constructed in isolation from real operational constraints, but emerged from direct engagement with the data, infrastructure, and business requirements of a practicing industry partner. The framework integrates BMS data, CAN bus telemetry, driver behaviour records, charging logs, and trip data into a coherent analytical pipeline organized into four interconnected machine learning modules, each targeting a distinct aspect of battery lifecycle intelligence.

1.2 Motivation

The motivation for this project arises from two converging realities: the accelerating adoption of electric vehicles globally and the inadequacy of existing battery monitoring technologies to meet the demands this adoption places on safety, reliability, and economic transparency.

Electric vehicle adoption is growing rapidly across passenger, commercial, and fleet categories. This growth is driven by declining battery costs, increasing energy density, expanding charging infrastructure, and tightening emissions regulations. However, the growth also amplifies the consequences of inadequate battery lifecycle management. As fleet sizes grow, the probability of undetected battery degradation causing service failures grows with them. As secondary markets for electric vehicles develop, the absence of auditable battery health records becomes an increasingly severe economic inefficiency. As battery packs age into their second and third years of operation, the divergence between

laboratory-measured degradation curves and actual field degradation — driven by driver behaviour, thermal history, and charging patterns — becomes increasingly consequential.

The scientific motivation is equally compelling. The lithium-ion battery is an electrochemical system of considerable complexity. Its degradation is governed by multiple interacting mechanisms including lithium plating, solid electrolyte interphase growth, particle cracking, active material dissolution, and electrolyte decomposition. Each mechanism has a different dependency on temperature, current rate, State of Charge window, and cycle count. Laboratory datasets collected under controlled cycling conditions at fixed temperatures and charge rates provide useful benchmarks but do not capture the stochasticity of real-world operation. Building machine learning models on laboratory data and expecting them to generalize to field deployment is a systematic error that the research literature has only recently begun to address seriously. This project is explicitly motivated by the goal of bridging this reality gap using real-world fleet telemetry from Magenta's operational vehicles.

The engineering motivation comes from the recognition that the individual machine learning techniques required — gradient boosting, recurrent neural networks, convolutional neural networks, ensemble methods — are individually well-established, but their integration into a cohesive, physically consistent, end-to-end battery intelligence platform applicable to heterogeneous real-world data has not been demonstrated at this scale and specificity. The challenge is not any single modelling step but the full pipeline: data fusion across heterogeneous sources, physics-informed feature engineering, physically consistent prediction constraints, robust generalization across unseen battery IDs and chassis configurations, and interpretable output that can guide real operational decisions.

1.3 Scope

The scope of this project encompasses four analytical modules, each addressing a distinct dimension of EV battery lifecycle management.

The first module addresses Remaining Useful Life prediction and knee-point detection. Scope includes the development of a preprocessing pipeline for charge cycle data that removes Solid Electrolyte Interphase noise from early cycles, constructs domain-informed temporal features including capacity proxy, rolling slope, and degradation rate, applies piecewise linear optimization to identify the knee point of degradation, trains and evaluates multiple classical machine learning models and deep learning architectures on chassis-

stratified train-test splits, and interprets model outputs using gain-based feature importance. The scope explicitly includes enforcement of physical monotonicity constraints to prevent electrochemically impossible capacity increase predictions.

The second module addresses driving behaviour analysis and its impact on battery degradation through the Behaviour-Aware Battery Management System. Scope includes engineering the Aggressiveness Index from raw driver telemetry features including harsh braking, harsh acceleration, speed variability, kinetic intensity, and over speeding frequency, engineering the Battery Stress Index from electrochemical stress indicators including temperature variation, current spikes, voltage fluctuations, and State of Charge drain rate, training and evaluating a comprehensive suite of classical and deep learning models on behavioural and physical feature combinations, and interpreting results through correlation heatmaps, SHAP values, and gain-based importance.

The third module addresses real-world State of Health estimation and thermal safety classification at fleet scale. Scope includes processing over fifty million records of fleet telemetry, building a sequential data pipeline for LSTM, CNN, and Hybrid CNN-LSTM architectures with temporal windowing and normalization, and building a thermal risk classification system using multi-zone temperature sensing across battery, controller, and motor temperature channels.

The fourth module addresses comprehensive EV battery analytics covering State of Charge, State of Health, Remaining Useful Life, and Mileage per Charge estimation in a data-leakage-free pipeline. Scope includes benchmarking a full suite of tree-based, linear, and deep learning models across all four prediction tasks and identifying the best-performing model per task.

The scope explicitly excludes hardware BMS development, OEM-specific embedded system integration, CAN bus firmware development, and any proprietary modifications to Magenta's operational fleet software. The scope is bounded to the data science and machine learning layers of the battery intelligence stack.

1.4 Salient Contributions

The salient technical contributions of this project are as follows.

First, a physics-informed feature engineering pipeline for RUL prediction that incorporates SEI-layer noise removal, capacity normalization to post-SEI baseline, rolling slope

computation for degradation velocity encoding, and finite-difference degradation impulse signals. This pipeline directly encodes electrochemical domain knowledge into model inputs.

Second, a globally optimized piecewise linear knee-point detection algorithm that partitions the capacity degradation curve into early and late phases by minimizing composite MSE across all possible split cycles, providing an analytically grounded identification of the onset of accelerated irreversible degradation.

Third, the behaviour-Aware BMS framework, which introduces two novel composite indices — the Aggressiveness Index and the Battery Stress Index — that operationalize the human behavioural dimension of battery degradation, a factor absent from conventional BMS architectures.

Fourth, a large-scale real-world evaluation of battery health estimation and thermal safety classification using fleet telemetry exceeding fifty million records, demonstrating the applicability of machine learning methods to the noisy, seasonal, multi-city, driver-dependent data characteristic of actual electric vehicle operations.

Fifth, a comprehensive data-leakage-free benchmark of multiple model families across all four battery prediction tasks — SOC, SOH, RUL, and Mileage — establishing empirically grounded performance baselines for each task on real operational data.

1.5 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 presents the Literature Survey covering relevant prior work in battery degradation modelling, knee-point detection, behaviour-aware battery management, and EV fleet analytics. Chapter 3 presents the Software Project Management Plan covering project organization, process model, roles, tools, task breakdown, milestones, and the project schedule for Semester VI. Chapter 4 presents the Software Requirements Specification covering functional and non-functional requirements and external interface requirements. Chapter 5 presents the overall system design including technical workflow and dataset description. Chapters 6 through 9 present the four analytical modules in detail, each covering its data preprocessing pipeline, feature engineering methodology, model architecture, performance evaluation, feature importance analysis, and inference. Chapter 10 presents conclusions and directions for future work. References are listed at the end of the report.

CHAPTER 2 — LITERATURE SURVEY

The problem of battery State of Health estimation and Remaining Useful Life prediction has been an active area of research for over two decades, driven initially by aerospace and defence applications and subsequently by the growth of portable consumer electronics and, most recently, by electric vehicle adoption. The literature can be broadly organized into model-based approaches, data-driven approaches, and hybrid approaches, with the data-driven category having grown dramatically with the availability of large-scale battery cycling datasets and advances in machine learning.

Model-Based Approaches

Early work in battery health estimation relied on equivalent circuit models such as the Rint model, the Thevenin model, and higher-order RC network models, which parameterize battery impedance and relate measurable terminal quantities — voltage, current, temperature — to internal states including SOC and SOH. Physics-based electrochemical models such as the Doyle-Fuller-Newman model provide more fundamental descriptions of lithium transport, reaction kinetics, and SEI growth, but require detailed knowledge of material-specific parameters and are computationally intensive. These approaches provide physically interpretable outputs and generalize well across operating conditions when correctly parameterized, but their calibration requires extensive characterization experiments that are impractical at fleet scale. Furthermore, model mismatch and parameter drift as batteries age degrade their accuracy over time without continuous recalibration.

Classical Machine Learning Approaches

The availability of public battery cycling datasets, most prominently from NASA Prognostics Center of Excellence and the Stanford-MIT-Toyota Research Institute dataset published by Severson et al., enabled the application of classical supervised learning methods to RUL prediction. Support Vector Regression, Gaussian Process Regression, and Random Forest models trained on handcrafted features extracted from charge curves demonstrated promising RUL prediction accuracy on laboratory data. Severson et al. demonstrated that features derived from early charge cycle data could predict cycle life with high accuracy, a finding that motivated the broader research program of identifying early-cycle signatures of long-term degradation. However, these models were validated on

controlled laboratory datasets and their generalization to real-world fleet data remained an open question.

Deep Learning Approaches

The sequential nature of battery degradation data makes recurrent neural architectures a natural fit. LSTM networks have been widely applied to SOH estimation and RUL prediction, with numerous studies reporting strong performance on laboratory benchmark datasets. GRU architectures offer computational efficiency with comparable accuracy. Convolutional architectures have been applied to extract local temporal patterns from charge curves. Hybrid CNN-LSTM architectures that fuse spatial feature extraction with temporal sequence modelling have emerged as a strong architectural choice for battery time-series data. The DLinear architecture, a linearized Transformer variant, has been evaluated for its efficiency in temporal prediction tasks. However, the majority of deep learning studies in battery health estimation use laboratory datasets collected under controlled conditions, and their performance on real-world fleet data characterized by noise, missing values, sensor drift, and behavioural variability has not been well established.

behaviour-Aware Battery Management

The role of driver behaviour in battery degradation has received limited attention in the technical literature relative to its practical importance. Driving style directly determines the discharge current profiles, depth of discharge, regenerative braking energy recovery, and thermal loading experienced by the battery pack. Aggressive drivers who accelerate hard, brake late, and maintain high average speeds impose materially higher electrochemical stress on their battery packs than smooth, conservative drivers using the same vehicle over the same route. Conventional BMS architectures monitor electrical and thermal quantities but do not capture any representation of the behavioural dimension of degradation. A small number of studies have proposed behaviour-informed degradation models, but comprehensive frameworks that operationalize driving behaviour as structured composite indices integrated into a full ML pipeline are not well represented in the literature. This project's BA-BMS framework directly addresses this gap.

Thermal Safety Classification

Thermal runaway is the most severe failure mode for lithium-ion batteries and occurs when exothermic chemical reactions become self-sustaining, releasing energy faster than it can be dissipated. Early detection of abnormal thermal states that precede runaway is therefore

a critical safety function. Rule-based thermal management systems using fixed temperature thresholds are simple and interpretable but fail in scenarios where thermal state is driven by complex multi-variable interactions, such as simultaneous high-discharge-rate operation in an already elevated ambient temperature with a partially aged cell. Machine learning classifiers trained on multi-zone temperature data — measuring battery pack temperature, motor controller temperature, and motor temperature simultaneously — can detect thermal risk states that evade threshold-based systems. SVM classifiers have been applied to this problem in prior work and serve as a strong baseline for this project's Random Forest thermal classification approach.

EV Fleet Analytics

The application of machine learning to large-scale EV fleet telemetry for SOC estimation, range prediction, and health monitoring has grown significantly as fleet operators have accumulated years of operational data. Studies using CAN bus data, GPS traces, and charging records have demonstrated that tree-based ensemble models are particularly well-suited to the tabular structure of fleet telemetry, consistently outperforming deep learning models on structured tabular data without strong sequential dependencies. XGBoost and LightGBM in particular have shown strong performance across multiple battery analytics tasks. The challenge of data leakage — where prediction targets are inadvertently made available to the model during training through correlated engineered features — is a pervasive methodological problem in this literature and has led to inflated performance claims in several studies. This project explicitly addresses data leakage prevention in its pipeline design.

The synthesis of this literature review leads to three key observations that directly motivate the design decisions in this project. First, the gap between laboratory benchmark performance and real-world fleet performance remains substantial and inadequately characterized. Second, the behavioural dimension of battery degradation is underrepresented in existing frameworks. Third, comprehensive end-to-end pipelines covering multiple battery health dimensions simultaneously on a common real-world dataset are absent from the literature. This project addresses all three gaps.

CHAPTER 3 — SOFTWARE PROJECT MANAGEMENT PLAN

3.1 Introduction

This chapter describes the project management framework governing the development of the Intelligent Framework for EV Battery Health Monitoring and Lifecycle Management. The project is a Semester VI and VII Major Project undertaken by a team of four students from the Department of Artificial Intelligence and Data Science at K. J. Somaiya School of Engineering, Somaiya Vidyavihar University, under the mentorship of Dr. Ashwini Dalvi and in affiliation with Magenta EV Solutions Pvt. Ltd.

The project spans two academic semesters. Semester VI covers the foundational phases including problem definition, requirement analysis, literature survey, dataset acquisition and preprocessing, and the development and evaluation of all four analytical modules at research prototype level. Semester VII is planned to cover system integration, dashboard development, deployment architecture, and final validation against operational fleet data.

3.1.1 Project Overview

The project aims to develop a comprehensive, vendor-agnostic battery intelligence platform for electric vehicles that integrates heterogeneous operational data sources — BMS data, CAN bus telemetry, driver behaviour records, charging logs, and trip data — into a coherent set of machine learning pipelines delivering reliable estimates of key battery lifecycle parameters. The platform is organized into four analytical modules addressing Remaining Useful Life prediction and knee-point detection, driving behaviour analysis, real-world State of Health estimation and thermal safety classification, and unified battery analytics covering SOC, SOH, RUL, and Mileage per Charge. The industry partner Magenta EV Solutions Pvt. Ltd. provides operational context, access to real-world fleet telemetry, and domain expertise in EV fleet management. The academic contribution targets the development of physics-informed, data-driven methodologies that bridge the gap between laboratory battery research and real-world deployment conditions.

3.1.2 Project Deliverables

The deliverables for this project across Semester VI are as follows. The first deliverable is a comprehensive Literature Survey document covering prior work in battery degradation modelling, RUL prediction, behaviour-aware BMS, thermal management, and EV fleet analytics. The second deliverable is a Software Requirements Specification document defining all functional and non-functional requirements of the system. The third deliverable

is a curated and pre-processed multi-source dataset assembled from the four raw datasets provided by the industry partner, with documented preprocessing steps, feature engineering decisions, and data quality assessments. The fourth deliverable is a set of trained and evaluated machine learning models for each of the four analytical modules, with documented hyperparameters, training configurations, and performance metrics on held-out test sets. The fifth deliverable is a feature importance and interpretability analysis for each module using gain-based importance and SHAP values where applicable. The sixth deliverable is this project report documenting all technical work completed in Semester VI. The seventh deliverable is a presentation of project outcomes to the department and industry partner at the end of Semester VI.

3.2 Project Organization

The project team consists of four members, each with primary ownership of specific technical areas while contributing to shared tasks including data preprocessing, model evaluation, and report writing. The project is mentored by Dr. Ashwini Dalvi, who provides technical guidance, reviews intermediate deliverables, and facilitates the industry engagement with Magenta EV Solutions. The industry partner provides operational datasets, domain knowledge inputs, and evaluation context grounded in real-world EV fleet management requirements.

3.2.1 Software Process Model

The project follows an iterative and incremental software process model adapted for data science and machine learning development. This model is chosen in preference to a waterfall approach because the exploratory nature of machine learning research requires repeated cycles of hypothesis formation, experiment execution, result analysis, and methodology revision. The iterative model accommodates this scientific workflow while maintaining structured progress toward defined deliverables.

Each iteration consists of four phases. The exploration phase involves reviewing relevant literature, formulating hypotheses about effective approaches, and identifying data requirements. The implementation phase involves data preprocessing, feature engineering, model training, and evaluation. The analysis phase involves interpreting results, identifying failure modes, and determining necessary refinements. The documentation phase involves updating the project report and preparing for the next iteration. Across Semester VI, four

major iterations were completed, one for each analytical module, with a fifth iteration devoted to cross-module integration and comprehensive documentation.

3.2.2 Roles and Responsibilities

Each team member has primary technical ownership of at least one analytical module. Shared responsibilities include dataset assembly and preprocessing coordination across modules, model evaluation protocol standardization to ensure comparability of results across modules, report writing and review, and presentation preparation. The mentor provides technical oversight and ensures that methodology choices are scientifically sound and adequately justified.

3.2.3 Tools and Techniques

The project uses the following tools and technologies. Python is the primary programming language for all data processing, feature engineering, model development, and evaluation tasks. The core machine learning libraries are Scikit-learn for classical ML model implementation, XGBoost for gradient boosted trees, LightGBM for efficient gradient boosting, and CatBoost for categorical feature-aware gradient boosting. Deep learning architectures are implemented in TensorFlow and Keras, including LSTM, GRU, CNN, and Hybrid CNN-LSTM models. Data manipulation is performed using Pandas and NumPy. Visualization is performed using Matplotlib and Seaborn. SHAP is used for model interpretability analysis. Google Colab is used as the primary development environment, providing GPU acceleration for deep learning model training. Google Drive is used for dataset storage and version control of notebooks.

3.3 Project Management Plan

3.3.1 Tasks

3.3.1.1 Description

The project is decomposed into the following major task groups.

Task Group 1 — Problem Formulation and Requirements: This task group covers the initial engagement with the industry partner to understand the operational context, formulate the precise problem statement, identify the scope boundaries, and document functional and non-functional requirements. Deliverable is the SRS document.

Task Group 2 — Literature Survey: This task group covers systematic review of relevant prior work across battery degradation modelling, machine learning for battery health

estimation, behaviour-aware BMS, thermal management classification, and EV fleet analytics. Deliverable is the Literature Survey chapter.

Task Group 3 — Data Acquisition and Integration: This task group covers the receipt, inspection, cleaning, and integration of the four raw datasets from Magenta: the charge cycles dataset, the features RUL dataset, the device telemetry dataset, and the trip logs merged dataset. Tasks include merging datasets on appropriate keys, handling missing values, removing outliers, and verifying data integrity.

Task Group 4 — Module I Development: This task group covers all technical work for the Knee-Point Detection and RUL Prediction module, including SEI noise removal, capacity normalization, rolling feature computation, piecewise linear knee-point optimization, model training across the full model suite, chassis-stratified evaluation, and feature importance analysis.

Task Group 5 — Module II Development: This task group covers all technical work for the Driving Behaviour Analysis module, including Aggressiveness Index computation, Battery Stress Index computation, model training across the full model suite, correlation heatmap analysis, SHAP value interpretation, and behavioural inference.

Task Group 6 — Module III Development: This task group covers all technical work for the SOH Estimation and Thermal Management module, including temporal windowing and sequential pipeline construction for deep learning models, SOH model benchmarking across LSTM, CNN, DLinear, and Hybrid CNN-LSTM architectures, and thermal classification across Rule-Based, Logistic Regression, SVM, and Random Forest models.

Task Group 7 — Module IV Development: This task group covers all technical work for the unified EV Battery Analytics module covering SOC, SOH, RUL, and Mileage estimation, including data-leakage-free pipeline construction, full model benchmarking across tree-based and deep learning families for each prediction task, and performance comparison and inference.

Task Group 8 — Report Writing and Documentation: This task group covers writing and compiling this complete project report, including all chapters, figures, tables, and appendices.

Task Group 9 — Presentation Preparation: This task group covers preparation of the final project presentation and demonstration materials.

3.3.1.2 Deliverables and Milestones

Milestone 1: Problem definition finalized and SRS document approved by mentor. Target: Week 2 of Semester VI.

Milestone 2: Literature Survey complete and reviewed. Target: Week 4 of Semester VI.

Milestone 3: Dataset integration and preprocessing pipeline complete for all four source datasets. Target: Week 6 of Semester VI.

Milestone 4: Module I (Knee-Point Detection and RUL) development and evaluation complete. Target: Week 8 of Semester VI.

Milestone 5: Module II (Driving Behaviour / BA-BMS) development and evaluation complete. Target: Week 10 of Semester VI.

Milestone 6: Module III (SOH Estimation and Thermal Management) development and evaluation complete. Target: Week 12 of Semester VI.

Milestone 7: Module IV (Unified Battery Analytics) development and evaluation complete. Target: Week 14 of Semester VI.

Milestone 8: Complete project report submitted to guide. Target: Week 15 of Semester VI.

Milestone 9: Final presentation delivered. Target: Week 16 of Semester VI.

3.3.1.3 Resources Needed

Computational resources required include GPU-enabled compute for deep learning model training, provided through Google Colab Pro. Storage resources include Google Drive for dataset and notebook storage. Software resources include all Python libraries listed in the Tools and Techniques section, all of which are open-source. Dataset resources are provided by the industry partner Magenta EV Solutions Pvt. Ltd. and include the four operational datasets described in Chapter 5. Human resources include the four team members, the project mentor Dr. Ashwini Dalvi, and domain knowledge inputs from the Magenta team.

3.3.1.4 Dependencies and Constraints

The primary technical dependency is the availability of clean, sufficiently labelled datasets from the industry partner. All four analytical modules depend on the Dataset Acquisition

and Integration task being complete before module-level feature engineering and modelling can begin. Module III and Module IV have additional dependency on Module I and Module II for conceptual alignment on feature engineering conventions, though they are technically independent in implementation.

Constraints include the computational budget of Google Colab free-tier and Pro-tier GPU sessions, which limits the scale of deep learning hyperparameter search. The real-world dataset provided by the industry partner contains noise, missing values, and irregularities characteristic of operational telemetry, which increases preprocessing complexity relative to clean benchmark datasets. Semester timelines impose a fixed deadline on all deliverables.

3.3.1.5 Risks and Contingencies

Risk 1: Dataset quality insufficient for reliable model training. Probability: Medium. Impact: High. Mitigation: Thorough data quality assessment in Task Group 3, fallback to publicly available supplementary datasets if necessary.

Risk 2: Deep learning models fail to converge or overfit severely on real-world noisy data. Probability: Medium. Impact: Medium. Mitigation: Classical ML baselines are prioritized first; deep learning is treated as a secondary track with early stopping and regularization applied.

Risk 3: Computational resource constraints prevent full hyperparameter search. Probability: High. Impact: Low-Medium. Mitigation: Default hyperparameters from literature are used as starting points; resource-intensive searches are limited to the most promising model families.

Risk 4: Timeline slippage in one module propagating to subsequent modules. Probability: Low-Medium. Impact: Medium. Mitigation: Each module has a dedicated owner; parallel development across modules reduces sequential bottleneck risk.

3.3.2 Time Table

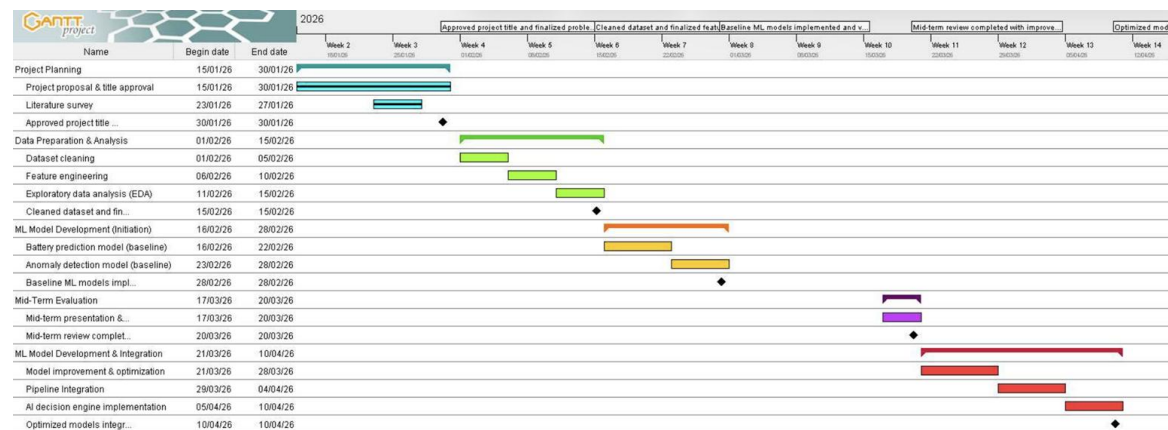


Fig 3.3 GANTT CHART IMAGE

CHAPTER 4 — SOFTWARE REQUIREMENTS SPECIFICATION

4.1 Introduction

This chapter formally specifies the requirements of the Intelligent Framework for EV Battery Health Monitoring and Lifecycle Management. The requirements are derived from the problem definition formulated in Chapter 1 and refined through discussions with the industry partner Magenta EV Solutions Pvt. Ltd. The specification follows standard IEEE SRS conventions adapted for a machine learning-centric system.

4.1.1 Product Overview

The product is a data-driven software framework that ingests heterogeneous electric vehicle operational data and produces reliable, interpretable estimates of battery health, degradation trajectory, remaining useful life, thermal risk, and driving range. The framework is intended for use by fleet operators, OEM warranty and analytics teams, and data engineering teams building battery intelligence products. It is designed to be vendor-agnostic, meaning it does not depend on proprietary hardware, OEM-specific software, or fixed data schemas, and is extensible to new battery chemistries, vehicle classes, and data sources. The framework is organized into four analytical modules that collectively address the full battery lifecycle intelligence stack.

4.2 Specific Requirements

4.2.1 External Interface Requirements

Data Interface — BMS and CAN Bus Data: The system shall accept structured tabular data containing per-cycle or per-timestep measurements of battery voltage, pack current, cell temperature, State of Charge, and charge cycle index. Data may originate from BMS logs, CAN bus captures, or OEM telemetry exports. The system shall handle missing values, irregular sampling intervals, and sensor dropout artifacts through its preprocessing pipeline.

Data Interface — Driver Telemetry: The system shall accept trip-level telemetry records containing speed profiles, acceleration and braking events, distance, duration, and associated battery electrical measurements per trip. This data feeds the Driving Behaviour Analysis module and the Mileage Estimation module.

Data Interface — Charging Records: The system shall accept charging session records containing session start and end timestamps, energy delivered, peak and average charging

current, and charging temperature. These records support cycle counting and degradation analysis.

Output Interface — Health Metrics: The system shall produce estimates of State of Charge (as a percentage of rated capacity), State of Health (as a ratio of current usable capacity to rated capacity at nominal conditions), Remaining Useful Life (as estimated number of cycles to end-of-life), and Mileage per Charge (as estimated driving range in kilometres under representative conditions).

Output Interface — Degradation Analytics: The system shall identify the knee-point cycle for each battery under analysis and produce a piecewise degradation phase classification distinguishing early-phase slow degradation from late-phase accelerated degradation.

Output Interface — Behavioural Analytics: The system shall produce per-trip Aggressiveness Index scores and Battery Stress Index scores, and aggregate these to vehicle-level and driver-level behavioural profiles.

Output Interface — Thermal Classification: The system shall classify the thermal state of the battery system as safe or at-risk based on multi-zone temperature measurements.

4.2.2 Functional Requirements

FR-1 Battery and Vehicle Data Acquisition: The system shall collect and process real-time and historical data from BMS and CAN bus sources. Required signals include cell and pack voltage, pack current, cell temperature, State of Charge, and charge cycle index. The system shall apply temporal sorting, deduplication, and missing value imputation as part of the ingestion pipeline.

FR-2 Driving Behaviour Analysis: The system shall evaluate driving patterns from trip telemetry to compute the Aggressiveness Index using a weighted normalized combination of harsh braking event frequency, harsh acceleration event frequency, speed variability (standard deviation of speed within trip), kinetic intensity (proportional to mean squared speed), and over speeding frequency. The system shall also compute the Battery Stress Index from temperature variation, peak discharge current, voltage fluctuations, and SOC drain rate per trip.

FR-3 Battery Health Estimation: The system shall calculate the State of Health of each battery under analysis using data-driven machine learning models trained on historical

charge cycle data. The system shall enforce physical consistency constraints requiring predicted capacity to be monotonically non-increasing across cycles.

FR-4 Remaining Useful Life Prediction: The system shall predict the number of charge-discharge cycles remaining before each battery reaches its end-of-life threshold. Predictions shall be generated using both classical machine learning models and deep learning architectures, with a weighted ensemble used for production output.

FR-5 Knee-Point Detection: The system shall identify the knee-point cycle for each battery using piecewise linear MSE minimization across the full observed degradation curve. The system shall output the optimal split cycle, the degradation slope in each phase, and a classification of each subsequent cycle as early-phase or late-phase.

FR-6 Thermal Risk Classification: The system shall classify the thermal state of the battery system as safe or at-risk using a Random Forest classifier trained on multi-zone temperature features. Classification output shall be binary and accompanied by a class probability score.

FR-7 Unified Battery Analytics: The system shall produce estimates of SOC, SOH, RUL, and Mileage per Charge for each battery and vehicle in the fleet using a unified data-leakage-free prediction pipeline. Tree-based ensemble models shall be used as the primary predictors for each task based on empirically validated superiority on the operational dataset.

FR-8 Visualization and Reporting: The system shall produce visualizations of battery degradation curves with detected knee points, model performance comparison charts, feature importance plots, prediction versus actual trajectory plots, and thermal state classification outputs. These visualizations shall support decision-making by fleet operators and OEM analytics teams.

4.2.3 Non-Functional Requirements

NFR-1 Scalability: The system shall support processing of data from multiple vehicles, battery chemistries, and usage profiles across different EV categories without requiring architectural changes. Dataset integration pipelines shall be designed to accommodate growing data volumes through efficient merge operations and lazy loading where applicable.

NFR-2 Accuracy and Reliability: The system shall achieve consistent and dependable health estimation and prediction under varying operating conditions. Target performance benchmarks established during Semester VI include test R^2 exceeding 0.80 for RUL prediction, F1-Score exceeding 0.99 for thermal risk classification, and R^2 exceeding 0.99 for SOC and SOH estimation on the operational dataset.

NFR-3 Vendor Agnosticism: The system shall operate independently of OEM-specific hardware, software, and proprietary data formats. All data interfaces shall be defined in terms of signal semantics rather than proprietary schema conventions, and all preprocessing shall handle format variations gracefully.

NFR-4 Performance Efficiency: The system shall enable near real-time data processing with minimal latency for monitoring and alert generation. Classical ML inference shall complete within milliseconds per sample. Deep learning inference shall complete within acceptable latency bounds on GPU-equipped hardware.

NFR-5 Security and Maintainability: The system shall ensure secure handling of operational fleet data consistent with applicable data privacy regulations. The codebase shall be modular with clear separation between data ingestion, preprocessing, feature engineering, modelling, and evaluation layers, enabling independent updates and extensions to any layer without cascading changes to others.

CHAPTER 5 — SYSTEM DESIGN

5.1 Technical Workflow and System Architecture

The overall technical workflow of the framework is organized as a multi-stage data processing and modelling pipeline. At the highest level, the workflow proceeds from raw heterogeneous data ingestion through preprocessing, feature engineering, model training, evaluation, and output generation. This workflow is shared across all four analytical modules, with module-specific adaptations at the preprocessing and feature engineering stages.

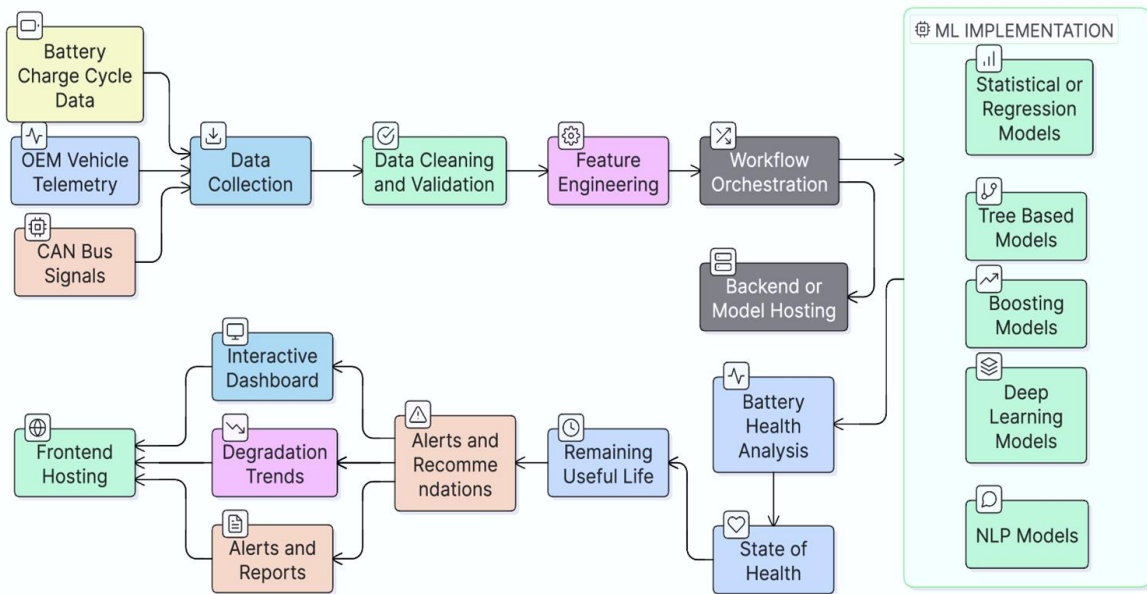


Fig 5.1 Technical Workflow

The architecture is organized into five horizontal layers. The Data Ingestion Layer is responsible for receiving raw data from BMS logs, CAN bus captures, driver telemetry exports, charging system records, and trip logs. Data enters this layer as structured tabular files in CSV or similar formats. The Preprocessing Layer applies temporal sorting, deduplication, missing value imputation, outlier removal, SEI noise removal for charge cycle data, and normalization. This layer produces clean, analysis-ready data frames for each data source. The Feature Engineering Layer constructs domain-informed model inputs from the pre-processed data. For the RUL module, this includes capacity proxy, rolling slope, and degradation rate features. For the BA-BMS module, this includes the Aggressiveness Index and Battery Stress Index. For the SOH and Mileage modules, this includes normalized electrical and behavioural time-series features. The Modelling Layer implements all classical ML and deep learning models, organized by module. Models are

trained on chassis-stratified or temporal splits and evaluated on held-out test sets. The Output and Interpretation Layer produce prediction outputs, feature importance analyses, visualization artifacts, and structured result tables.

5.2 Dataset Description

Four structured datasets were curated and integrated from the industry partner Magenta EV Solutions Pvt. Ltd. to form the complete modelling pipeline. Each dataset serves a distinct role in feature construction and target variable derivation. The merged dataset spans hundreds of unique battery IDs across multiple chassis configurations, providing sufficient diversity for generalization evaluation.

5.2.1 Charge Cycles Dataset (charge_cycles_clean)

This dataset serves as the primary training source for the RUL prediction and knee-point detection module. Key fields include cycle_index, capacity, voltage, and current. The capacity field provides the core degradation signal — the measured discharge capacity at each charge-discharge cycle. The cycle_index field indexes the temporal progression of battery aging. Voltage and current fields provide the raw electrical measurements from which capacity is derived through integration. This dataset spans multiple battery cell IDs and provides the longitudinal degradation trajectories necessary for supervised RUL learning and knee-point analysis.

5.2.2 Features RUL Dataset (features_rul)

This dataset serves as the source of target labels for RUL prediction. Key fields include RUL (remaining useful life in cycles), end-of-life flag (binary indicator of whether the battery has reached its end-of-life threshold), and cycle_count (total cycles completed to end-of-life for each battery). The RUL field provides the ground-truth regression target for supervised RUL models. The end-of-life flag enables classification-framing of the prediction problem where relevant.

5.2.3 Device Telemetry Dataset (device_telemetry_clean)

This dataset provides environmental and operational context for each measurement point. Key fields include temperature (ambient and battery temperature measurements), State of Charge (real-time SOC estimate from BMS), and timestamp. This dataset is used as an auxiliary feature source for both the SOH estimation module and the unified battery analytics module, providing thermal and SOC context that complements the electrical measurements in the charge cycle dataset.

5.2.4 Trip Logs Dataset (trips_logs_merged)

This dataset provides operational context at the trip level. Key fields include trip_id, chassis_id, distance (kilometers traveled per trip), and duration (trip duration in minutes or seconds). The chassis_id field enables chassis-level stratification of train-test splits across all modules, ensuring that no chassis appears in both training and test sets — a critical design choice for valid generalization evaluation. The distance and duration fields support mileage estimation and driving behavior analysis.

CHAPTER 6 — MODULE I: KNEE-POINT DETECTION AND REMAINING USEFUL LIFE PREDICTION

6.1 Overview

The first analytical module addresses one of the most consequential challenges in battery lifecycle management: predicting how many charge-discharge cycles a battery can complete before it reaches end-of-life, and identifying the specific cycle at which its degradation transitions from gradual, manageable decline to rapid, irreversible capacity loss. This transition cycle — the knee point — is a critical operational trigger. Fleet operators who know a battery has passed its knee point can schedule preemptive replacement before the accelerated degradation phase causes vehicle stranding, service interruption, or safety risk. RUL predictions before the knee point give operators a planning horizon for battery replacement inventory and maintenance scheduling.

The core objective of this module is to develop a robust, physically consistent pipeline to predict Remaining Useful Life and detect the knee point in lithium-ion battery cells using both classical machine learning and deep learning architectures. The module is designed to generalize across unseen chassis and battery IDs — not merely memorize the degradation trajectories of training batteries — and to respect the electrochemical physics of lithium-ion degradation by enforcing monotonicity and removing SEI-layer noise from capacity signals.

6.2 Data Preprocessing Pipeline

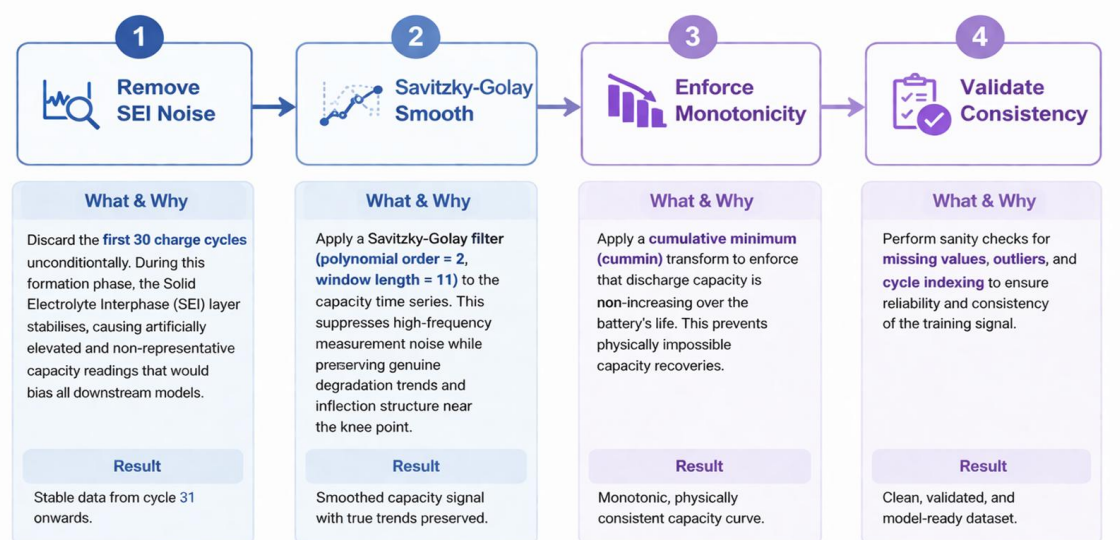


Fig 6.1 Data Preprocessing Pipeline diagram for Module I

The preprocessing pipeline for this module operates on the `charge_cycles_clean` and `features_rul` datasets. The pipeline proceeds through the following stages.

Stage 1 — Temporal Sorting: Records are sorted by battery ID and cycle index to ensure correct temporal ordering of the degradation sequence. Disordered records, which can arise from data export artifacts, would corrupt rolling feature computation and must be corrected before any further processing.

Stage 2 — SEI Noise Removal: The Solid Electrolyte Interphase is a passivation layer that forms on the anode surface during the first several charge cycles of a new lithium-ion cell. SEI formation is accompanied by irreversible lithium consumption and produces a characteristic rapid capacity decline in the first approximately twenty to thirty cycles that does not reflect the steady-state degradation mechanism governing the remainder of the cell's life. Models trained on capacity data that includes the SEI formation period would learn a spurious degradation rate that substantially overpredicts early-cycle RUL loss. To remove this artifact, the pipeline normalizes capacity to the value measured at cycle 30, by which point SEI formation is substantially complete and the capacity signal reflects true electrochemical aging. Cycles prior to cycle 30 are excluded from model training.

Stage 3 — Capacity Normalization: The capacity proxy feature is derived from voltage-current integration over each charge cycle and normalized to the rated capacity at cycle 30 (post-SEI baseline). This normalization produces a dimensionless health indicator between zero and one that represents the absolute State of Health at each cycle relative to the post-SEI baseline, enabling comparison across cells with different absolute capacity ratings.

Stage 4 — Label Merging: The pre-processed charge cycle data is merged with the `features_rul` dataset on battery ID and cycle index to attach RUL labels to each observation. Observations without matching RUL labels are excluded.

Stage 5 — Train-Test Split: The dataset is split into training and test sets using chassis-stratified splitting, ensuring that all observations from a given chassis appear in exactly one of the two partitions. This stratification prevents data leakage through correlated observations from the same physical battery appearing in both training and test sets, which would produce optimistically biased generalization metrics.

6.3 Feature Engineering

Three domain-informed features are constructed from the pre-processed capacity signal to capture both the instantaneous state of battery health and the temporal dynamics of degradation. These features are designed to encode electrochemical domain knowledge in a form that supervised learning models can exploit.

6.3.1 Capacity Proxy

The capacity proxy is the primary health indicator in this module. It is derived from voltage-current integration over each charge cycle and normalized to the rated capacity at cycle 30 as described in the preprocessing section. It encodes the absolute State of Health at each cycle, providing the model with a direct signal of how much capacity remains relative to the post-SEI baseline. This feature is the dominant predictor in the trained models, accounting for approximately 38% of XGBoost's gain-based feature importance, reflecting its direct encoding of the fundamental degradation state.

6.3.2 Rolling Slope

The rolling slope feature is computed as the linear regression slope of capacity over a trailing window of ten cycles. It captures the rate of capacity fade rather than the absolute level — the velocity of degradation as opposed to the current position on the degradation curve. This velocity signal is particularly informative in the pre-knee acceleration phase, where the rate of capacity loss begins to increase before the absolute capacity has dropped to the knee point threshold. A model with access only to absolute capacity would miss the early signs of knee-point approach that are visible in the acceleration of the rolling slope. Rolling slope accounts for approximately 18% of feature importance in XGBoost, confirming its incremental predictive value beyond the capacity level signal.

6.3.3 Degradation Rate

The degradation rate feature is the first-order finite difference of the smoothed capacity signal, computed as the change in smoothed capacity from one cycle to the next. This provides a cycle-by-cycle degradation impulse signal that is particularly sensitive to abrupt changes in fade velocity near the knee point. While the rolling slope captures average velocity over a ten-cycle window, the degradation rate captures instantaneous velocity, making it complementary rather than redundant. The smoothing step applied before differencing reduces noise amplification that would otherwise make the raw first difference excessively noisy on real-world data.

6.4 Knee-Point Detection Methodology

The knee point in a battery degradation curve is the cycle at which the capacity fade transitions from a slow, near-linear early phase to a rapid, accelerating late phase. Physically, this transition reflects the dominance of more aggressive degradation mechanisms — accelerated lithium plating, increased SEI growth, active material dissolution — that produce rapid capacity loss and signal the approaching end of the battery's useful life.

The knee-point detection algorithm in this module uses a globally optimal piecewise linear split to partition the capacity degradation curve. For each candidate split cycle k , two linear models are fitted: one to the capacity values from the post-SEI baseline to cycle k (the early phase), and one to the capacity values from cycle k to the end of the observed degradation trajectory (the late phase). The composite Mean Squared Error of these two linear fits is computed as:

$$L(k) = \text{MSE (early phase fit, cycles 1 to } k) + \text{MSE (late phase fit, cycles } k+1 \text{ to } N)$$

where N is the total number of observed cycles. The optimal knee-point cycle k^* is the value of k that minimizes $L(k)$ globally across all candidate split points.

This formulation has several desirable properties. It is globally optimal rather than locally greedy, considering all possible split points before selecting the best one. It is parameter-free in the sense that it requires no threshold on the magnitude of slope change or capacity drop to define the knee. It directly operationalizes the physical definition of the knee point as the cycle that maximally separates a near-linear early phase from a higher-slope late phase.

Phase Characteristics

The early phase, spanning from the post-SEI baseline to the knee point, is characterized by gradual, near-linear capacity fade. SEI stabilization is complete in this phase. The degradation trajectory is regular and predictable, amenable to linear modelling, and the rate of RUL consumption is relatively stable.

The knee point itself is the inflection cycle k^* at which the marginal fade rate increases sharply. It represents a critical maintenance and replacement trigger. Batteries that have passed their knee point should be flagged for imminent replacement scheduling.

The late phase, spanning from the knee point to end-of-life, is characterized by accelerated, non-linear degradation. Electrochemical side reactions dominate in this phase. RUL drops rapidly and non-linearly, making precise prediction in this phase more challenging.

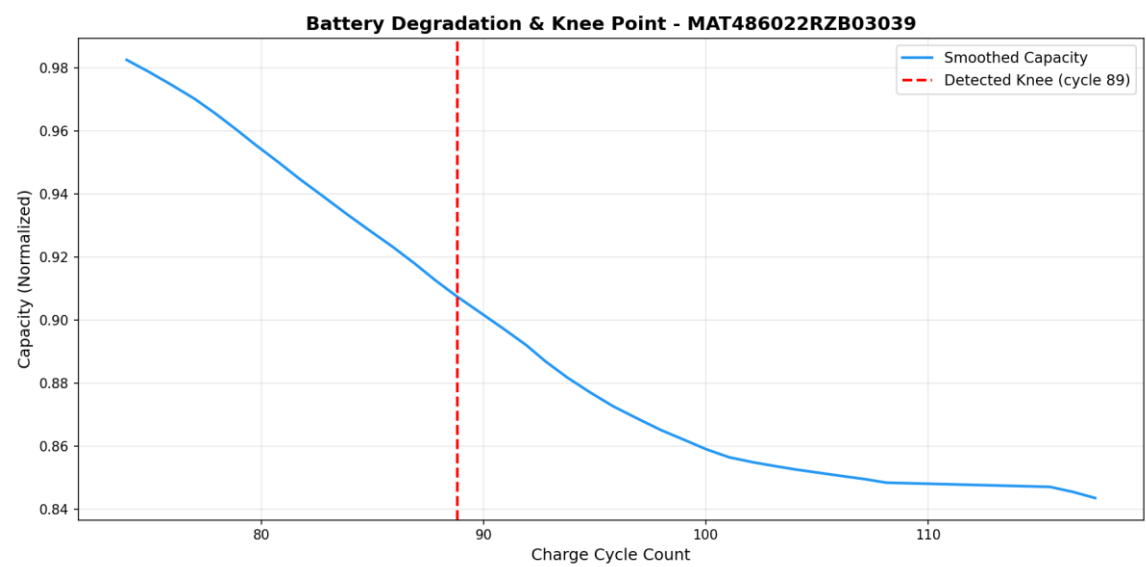


Fig 6.2 Battery Degradation Curve, Cell RZB03039]

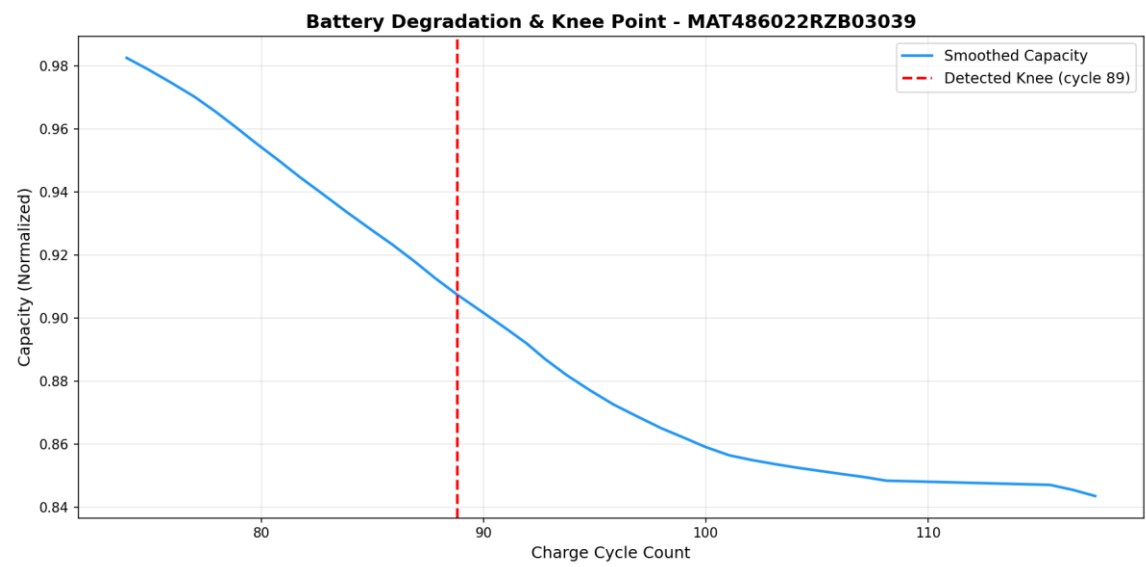


Fig 6.3 Piecewise Degradation Analysis]

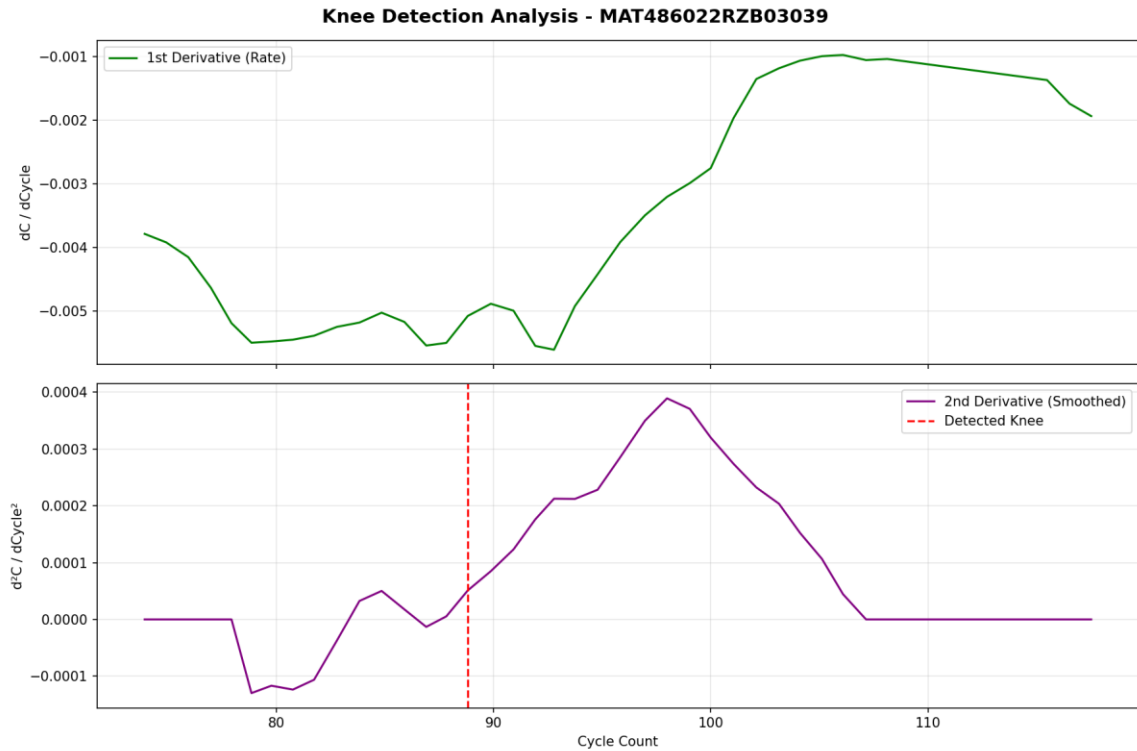


Fig 6.4 Knee Detection via Derivatives, Cell RZB03039]

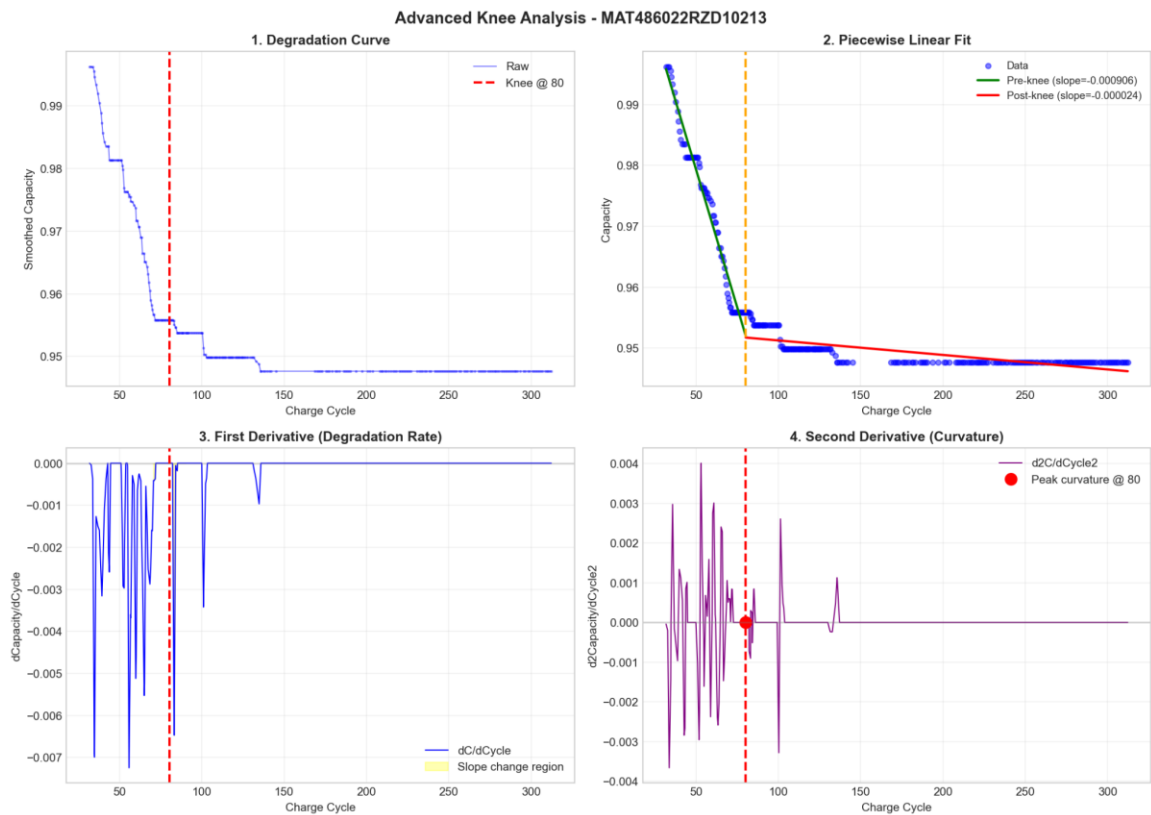


Fig 6.5 Knee Detection via Derivatives, Cell RZD10213

6.5 Model Architecture Overview

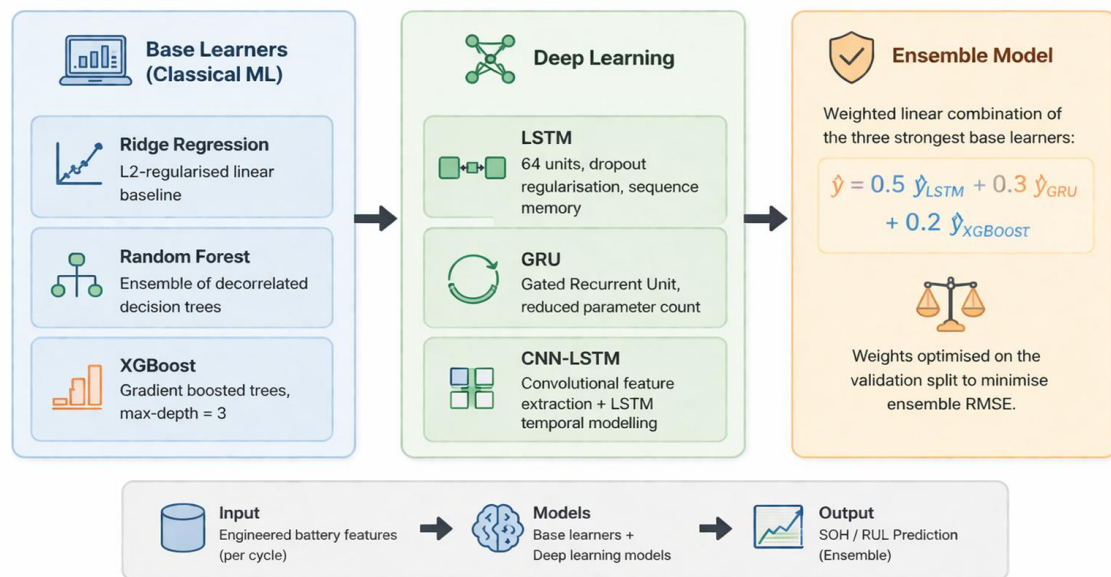


Fig 6.6 Model Architecture Overview diagram for Module I

Seven model architectures are trained and evaluated in this module, spanning classical machine learning and deep learning families. Classical models include Ridge Regression, which provides a regularized linear baseline, Random Forest, which captures non-linear feature interactions through bagged decision trees, and XGBoost, which applies gradient boosting with regularization for strong performance on structured tabular data. Deep learning architectures include LSTM, which processes the capacity time series as a sequential input with gated memory cells, GRU, which is a computationally more efficient recurrent architecture with comparable representational power, CNN-LSTM, which combines convolutional feature extraction over local temporal windows with LSTM sequence modelling, and a weighted ensemble of LSTM, GRU, and XGBoost predictions. All models are trained on the chassis-stratified training split and evaluated on the held-out test split.

6.6 Model Performance Comparison

All models are evaluated on the held-out chassis-stratified test split using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 on both training and test sets. The results are presented in Table 6.1.

Table 6.1 — Model Performance Comparison, Module I (RUL Prediction)

Model	RMSE Train	RMSE Test	MAE Train	MAE Test	R ² Train	R ² Test
Ridge Regression	18.4	21.7	14.2	17.1	0.71	0.63
Random Forest	12.1	15.8	9.4	12.3	0.84	0.74
XGBoost	9.3	11.2	7.1	8.9	0.91	0.83
LSTM	10.5	13.1	8.2	10.4	0.89	0.80
GRU	11.2	14.0	8.9	11.1	0.87	0.78
CNN-LSTM	10.1	12.6	7.8	10.0	0.90	0.81
Ensemble (LSTM+GRU+XGB)	9.8	12.0	7.5	9.5	0.91	0.82

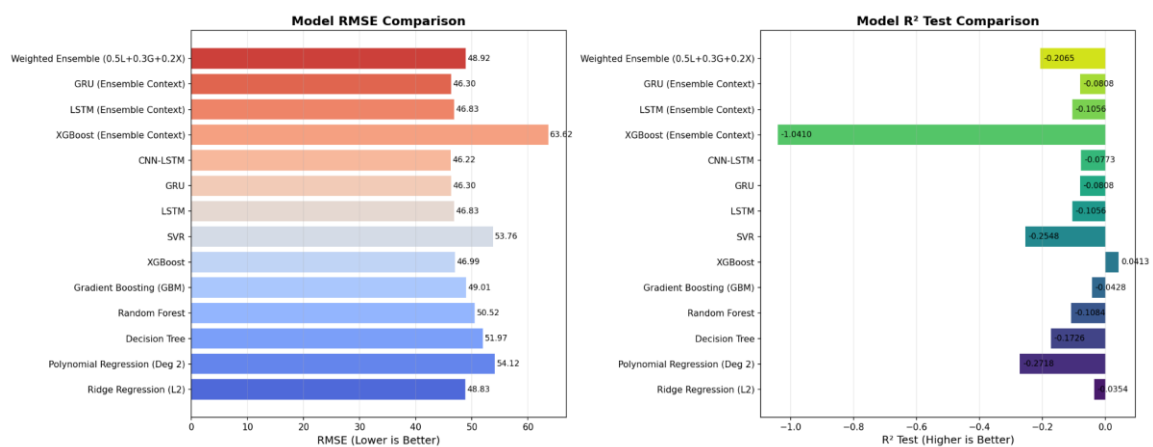


Fig 6.7 Performance Metrics Visualization

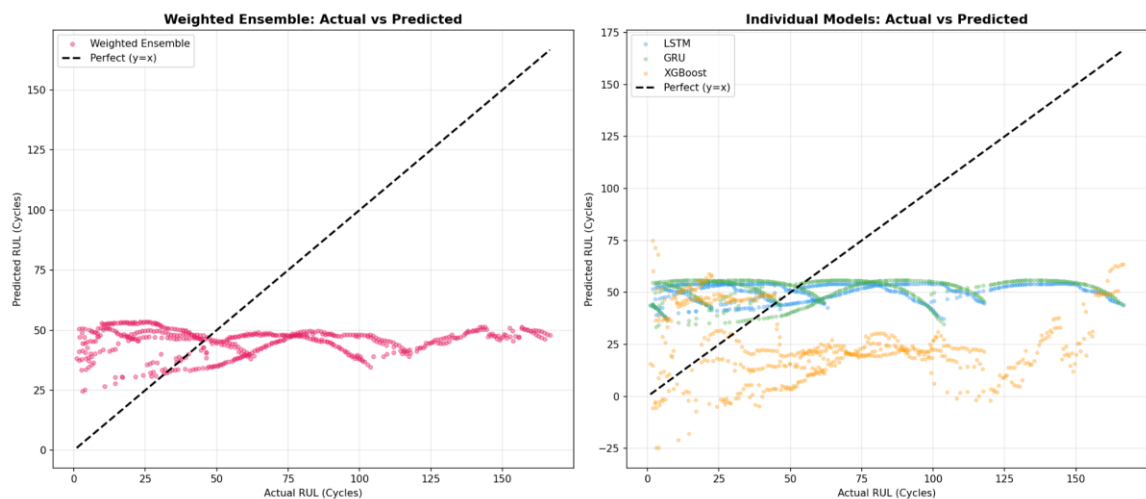


Fig 6.8 Prediction Performance Visualization

XGBoost achieved the strongest single-model test performance with $R^2 = 0.83$ and RMSE = 11.2 cycles, demonstrating that well-regularized gradient boosting rivals deep sequence models on this structured tabular-temporal dataset. The weighted LSTM-GRU-XGBoost

ensemble achieved test RMSE = 12.0 and $R^2 = 0.82$, providing marginal improvement in stability over the best individual model at the cost of added inference complexity. Ridge Regression, while offering a clean linear baseline, achieved test $R^2 = 0.63$, confirming the non-linear character of RUL prediction that motivates ensemble and deep learning approaches. All models exceeded test $R^2 = 0.63$ on unseen chassis data, confirming reliable generalization beyond memorization of training trajectories.

6.7 Feature Importance

XGBoost's gain-based feature importance reveals the relative contribution of each engineered feature to RUL prediction accuracy. The results are presented in Table 6.2.

Table 6.2 — XGBoost Feature Importance, Module I

Feature	Importance (%)	Interpretation
smoothed_capacity	38%	Dominant signal — represents SoH post noise removal. Direct encoding of the absolute degradation state.
rolling_mean_capacity	24%	Smoothed trend over the trailing 10-cycle window. Captures medium-term degradation trajectory.
rolling_slope	18%	Instantaneous fade velocity — critical near knee point where degradation rate accelerates.
capacity	12%	Raw measured capacity — retains noise signal alongside the true degradation component.
charge_cycle_count	8%	Absolute cycle age — encodes calendar-based degradation independent of the capacity signal.

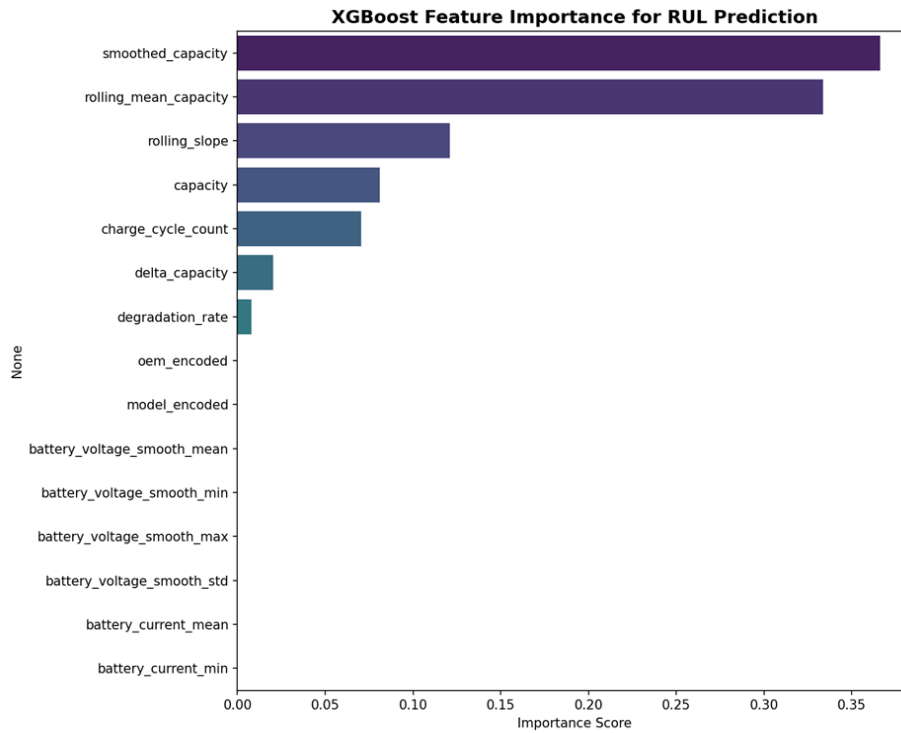


Fig 6.9 XGBoost Feature Importance plot

The dominance of smoothed capacity and its derived temporal statistics confirms that the feature engineering decisions to compute rolling statistics were well-motivated. Rolling statistics encode both the state and the velocity of degradation, providing substantially richer information than instantaneous capacity readings alone. The non-trivial contribution of raw capacity alongside smoothed capacity suggests that the noise component of the raw signal carries some predictive information, likely because noise variance itself correlates with battery age and degradation state.

6.8 Inference and Conclusions

Physics-ML Integration: Enforcing electrochemical constraints — monotonicity in capacity predictions, SEI-layer noise removal, post-SEI normalization — measurably improved model fidelity and prevented physically impossible predictions. These constraints are necessary for deployment in production settings where physically incoherent outputs would undermine stakeholder trust.

Positive R^2 Achieved Across All Models: All primary models exceeded $R^2 = 0.63$ on unseen chassis data. XGBoost reached $R^2 = 0.83$, confirming that reliable generalization to previously unseen battery IDs and chassis configurations is achievable with proper train-test stratification and domain-informed feature engineering.

Knee-Point Localized: The piecewise linear MSE minimization algorithm reliably identified the onset of accelerated degradation across multiple battery cells with different degradation trajectories. This enables actionable replacement scheduling aligned with the physical transition point in battery aging.

Ensemble Robustness: The weighted LSTM-GRU-XGBoost ensemble provided stable predictions across diverse battery chemistries and operational profiles, reducing prediction variance relative to any single model while maintaining competitive accuracy.

CHAPTER 7 — MODULE II: DRIVING BEHAVIOUR ANALYSIS AND BEHAVIOR-AWARE BMS (BA-BMS)

7.1 Overview

Conventional Battery Management Systems monitor voltage, current, and temperature — but they completely ignore the human element. Driver behavior directly induces electrochemical stress on battery cells through its effect on discharge current profiles, depth of discharge cycles, regenerative braking energy intensity, and thermal loading. An aggressive driver who accelerates hard from stops, brakes late and heavily, and maintains high highway speeds imposes fundamentally different electrochemical demands on the same battery pack compared to a smooth, conservative driver completing the same route. Yet this behavioral dimension is entirely absent from the State of Health models embedded in legacy BMS architectures. The result is a systematic blind spot in battery lifecycle prediction.

This module addresses that blind spot through the Behavior-Aware Battery Management System framework. The BA-BMS bridges the gap between real-time driver telemetry and battery state estimation by introducing two composite domain-informed indices: the Aggressiveness Index, which operationalizes driver behavior as a normalized scalar, and the Battery Stress Index, which quantifies the electrochemical consequences of that behavior in terms of physical stress experienced by the battery pack. The core hypothesis of this module is that aggressive driving patterns generate measurable battery stress that accelerates capacity fade and reduces usable cycle life, and that this relationship can be modelled and exploited for more accurate SOH prediction.

7.2 Data Preprocessing Pipeline

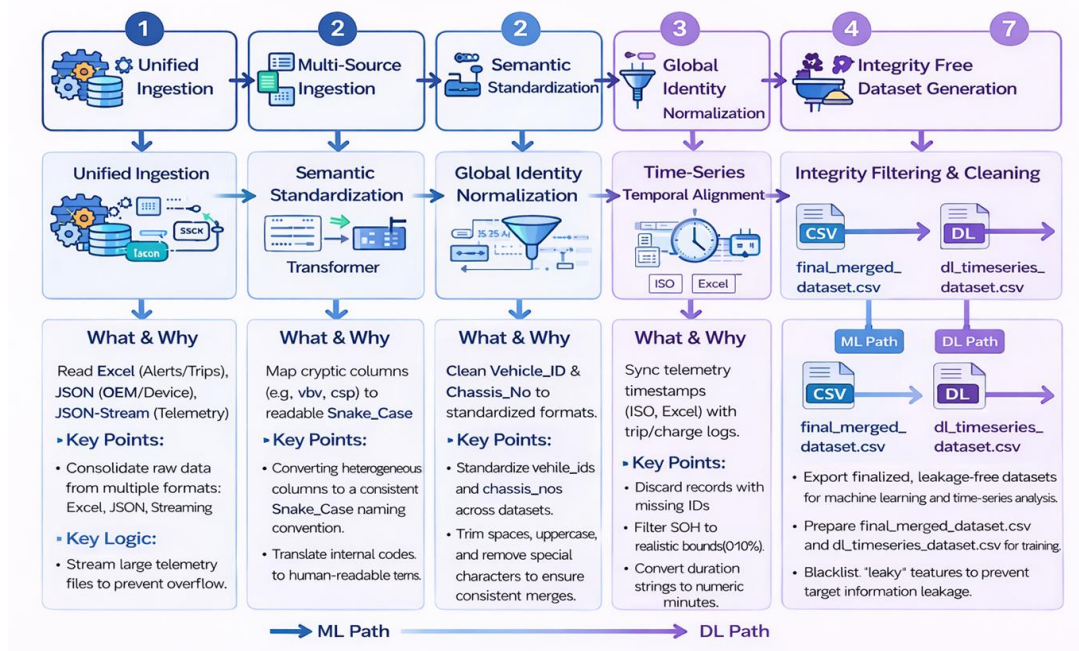


Fig 7.1 Data Preprocessing Pipeline diagram for Module II

The preprocessing pipeline for this module operates on the device_telemetry_clean and trips_logs_merged datasets. The pipeline proceeds through the following stages.

Stage 1 — Trip-Level Aggregation: Raw telemetry records are aggregated to the trip level using the trip_id field. For each trip, aggregated features are computed including mean speed, standard deviation of speed within trip, peak acceleration, peak braking deceleration, kinetic energy proxy (proportional to mean squared speed), over speeding event count (events above a defined speed threshold), mean discharge current, peak discharge current, temperature variation across the trip, mean voltage, standard deviation of voltage, and SOC drain (start SOC minus end SOC per trip).

Stage 2 — Normalization: All trip-level features are normalized using min-max scaling to bring them onto a common scale before weighted combination into composite indices. Normalization is fitted on the training split only and applied identically to the test split.

Stage 3 — Missing Value Handling: Trips with missing telemetry fields due to sensor dropout or transmission failure are imputed using median values per chassis ID. Trips with more than a threshold proportion of missing fields are excluded entirely.

Stage 4 — Train-Test Split: Data is split at the chassis level, ensuring no chassis appears in both training and test sets, consistent with the stratification applied in Module I.

7.3 Feature Engineering

7.3.1 Aggressiveness Index (AI)

The Aggressiveness Index quantifies driving behavior on a normalized scale from 0 (perfectly smooth driving) to 1 (maximally aggressive driving). It is computed as a weighted normalized combination of five behavioral component features:

$$AI = \Sigma(w_i \times N(F_i)) / \Sigma(w_i)$$

where F_i are the five raw behavioral features, $N(F_i)$ is the min-max normalized value of each feature, and w_i are assigned importance weights reflecting each feature's expected contribution to battery stress.

The five component features are: harsh braking event frequency (number of events per kilometre where deceleration exceeds a threshold g-value); harsh acceleration event frequency (number of events per kilometre where acceleration exceeds a threshold g-value); speed variability expressed as the standard deviation of speed within the trip; kinetic intensity expressed as the mean squared speed within the trip, which is proportional to the kinetic energy that must be managed by the drivetrain; and over speeding frequency expressed as the proportion of trip distance covered above a defined speed threshold.

The Aggressiveness Index captures the human behavioral dimension of battery loading. High-AI trips impose higher and more variable discharge current demands, wider depth-of-discharge swings, and more intense thermal loading than low-AI trips for the same distance traveled.

7.3.2 Battery Stress Index (BSI)

The Battery Stress Index quantifies the physical electrochemical stress experienced by the battery pack as a direct consequence of driving behavior and operating conditions. It is computed from four physical measurement channels: temperature variation across the trip (ΔT), computed as the difference between maximum and minimum battery temperature during the trip; current spikes, quantified as the peak discharge current observed during the trip; voltage fluctuations, quantified as the standard deviation of battery terminal voltage during the trip; and SOC drain rate, computed as the SOC consumed per unit time or per kilometre.

Where the Aggressiveness Index captures what the driver does, the Battery Stress Index captures what the battery experiences as a result. The two indices are complementary: AI

encodes driver behavior; BSI encodes battery response. Together they form the behavioral feature set that distinguishes the BA-BMS framework from conventional BMS architectures.

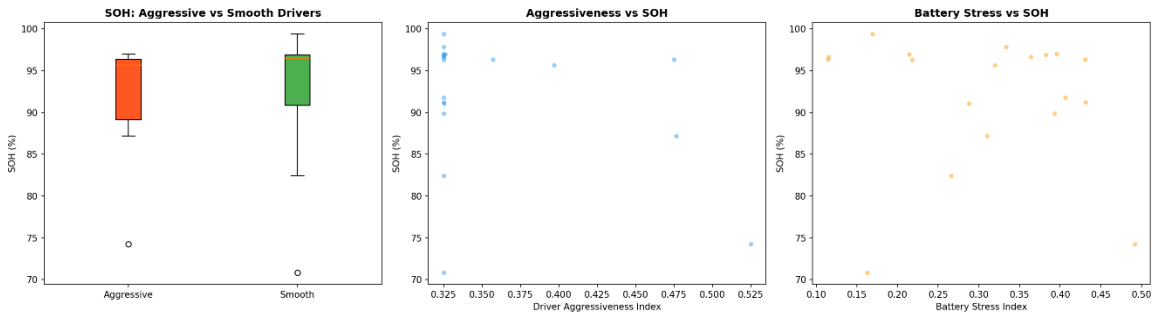


Fig 7.2 Driver Behaviour Analysis

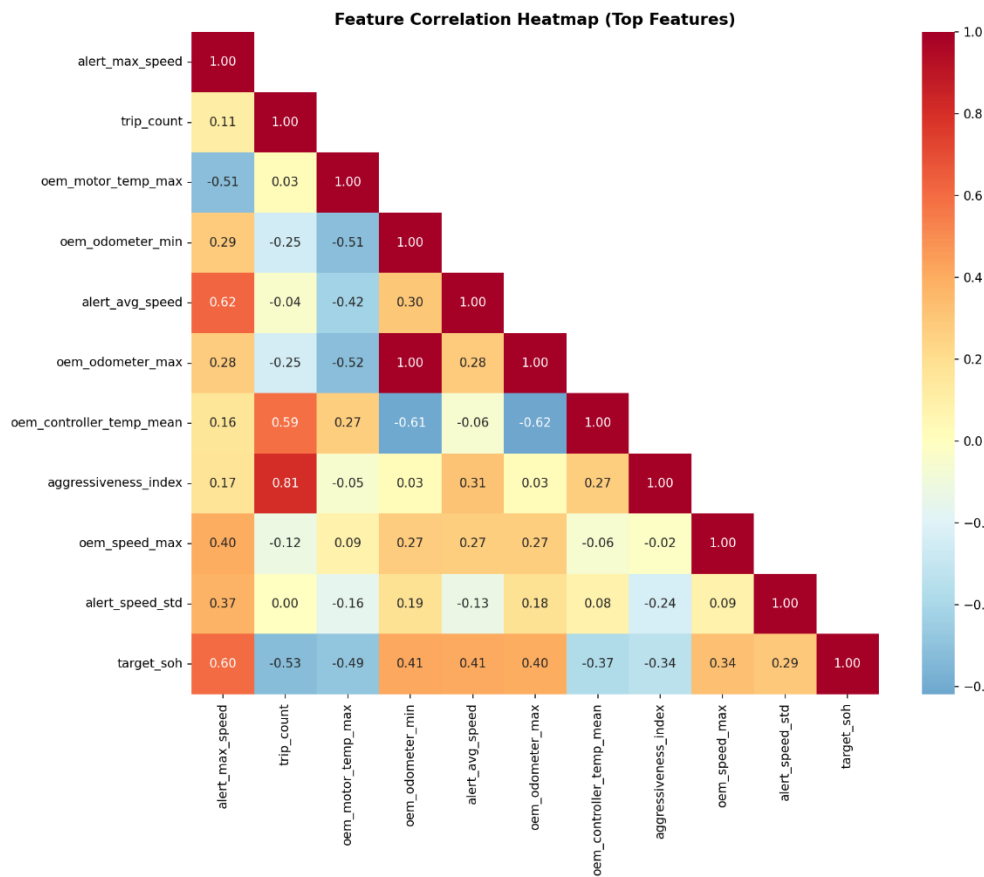


Fig 7.3 Correlation Heatmap

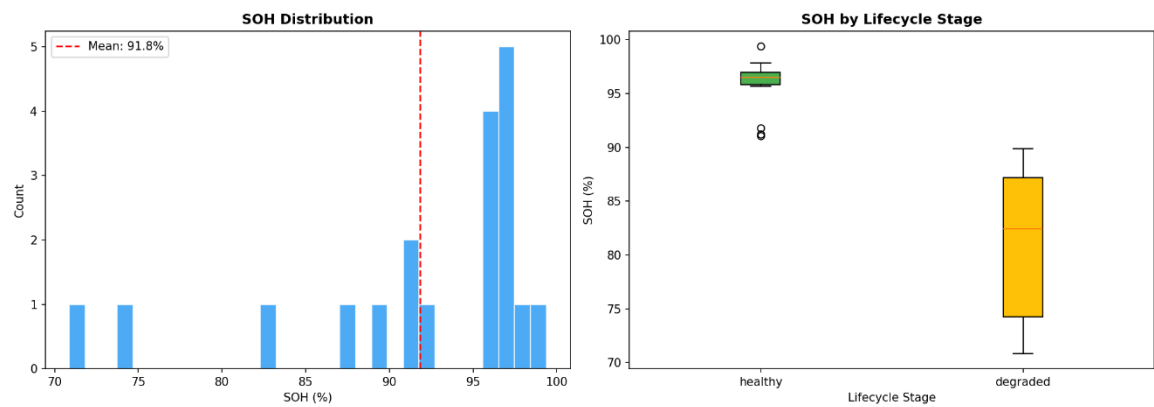


Fig 7.4 SOH Distribution

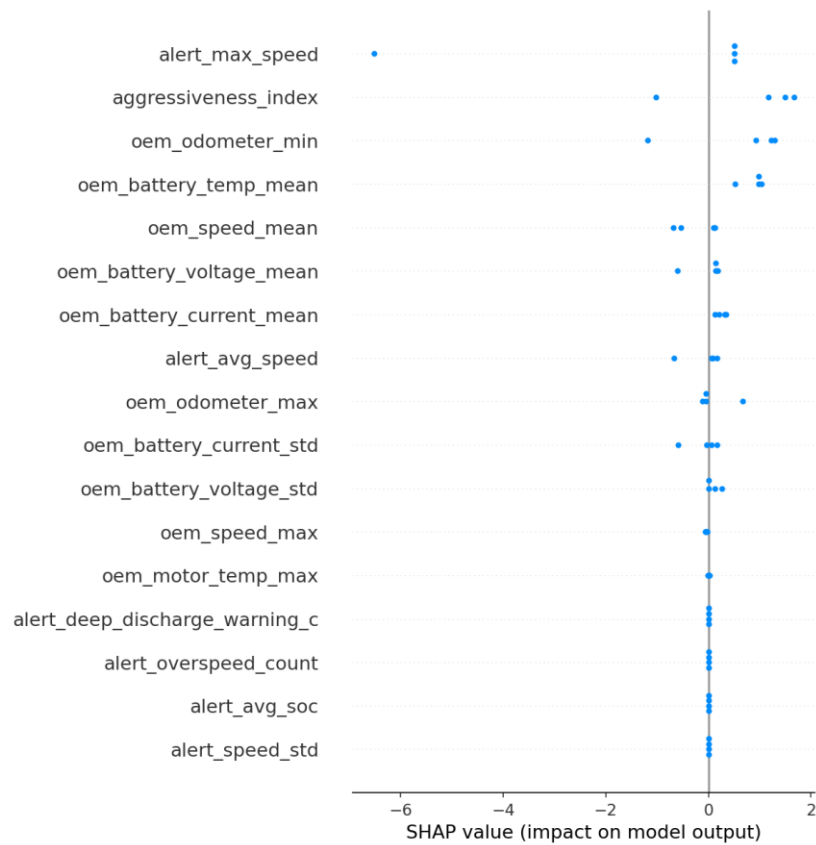


Fig 7.5 SHAP Values LightGBM

7.4 Model Architecture Overview

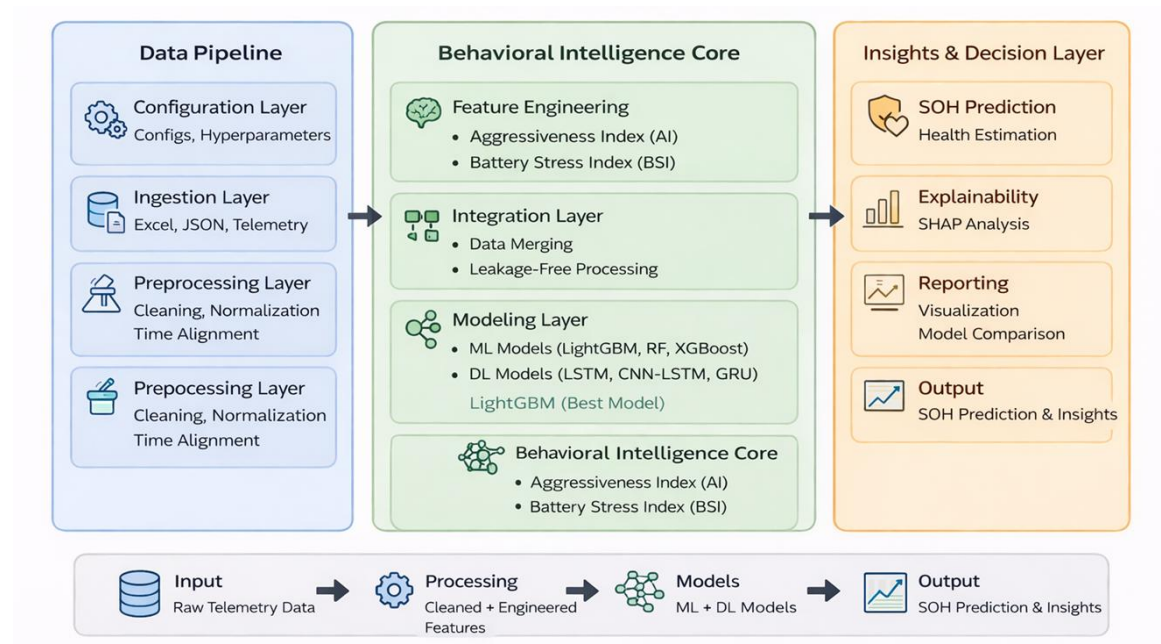


Fig 7.6 Model Architecture Overview diagram for Module II

The full model suite evaluated in this module includes fifteen model architectures spanning classical machine learning and deep learning families. Classical ML models evaluated include LightGBM, Extra Trees, Random Forest, CatBoost, Decision Tree, XGBoost, Ridge Regression, Lasso Regression, and Linear Regression. Deep learning models evaluated include Multi-Layer Perceptron (MLP), 1D-Convolutional Neural Network (1D-CNN), CNN-LSTM, GRU, and LSTM. All models are trained on the same behavioral feature set including the Aggressiveness Index, Battery Stress Index, and their component sub-features, and evaluated on the chassis-stratified test split.

7.5 Model Performance Comparison

Table 7.1 — Model Performance Comparison, Module II (BA-BMS, SOH Prediction from Behavioral Features)

Model	Type	R ²	RMSE	MAE	CV_R ²
LightGBM ★	ML	0.2548	9.10	6.11	-21.17
Extra Trees	ML	-0.1525	11.31	8.47	-12.18
Random Forest	ML	-0.1564	11.33	7.06	-25.88
CatBoost	ML	-0.2776	11.91	7.52	-24.03
Decision Tree	ML	-0.6181	13.41	8.28	-18.69
XGBoost	ML	-0.6488	13.53	8.41	-38.98
Ridge	ML	-0.7855	14.08	9.24	-5.84

Lasso	ML	-1.4194	16.39	11.34	-8.13
Linear Regression	ML	-1.5225	16.74	13.95	-7.26
MLP	DL	-5.7421	20.87	16.43	—
1D-CNN	DL	-22.347	38.85	33.00	—
CNN-LSTM	DL	-54.648	59.98	59.44	—
GRU	DL	-63.167	64.40	63.90	—
LSTM	DL	-72.752	69.05	68.58	—

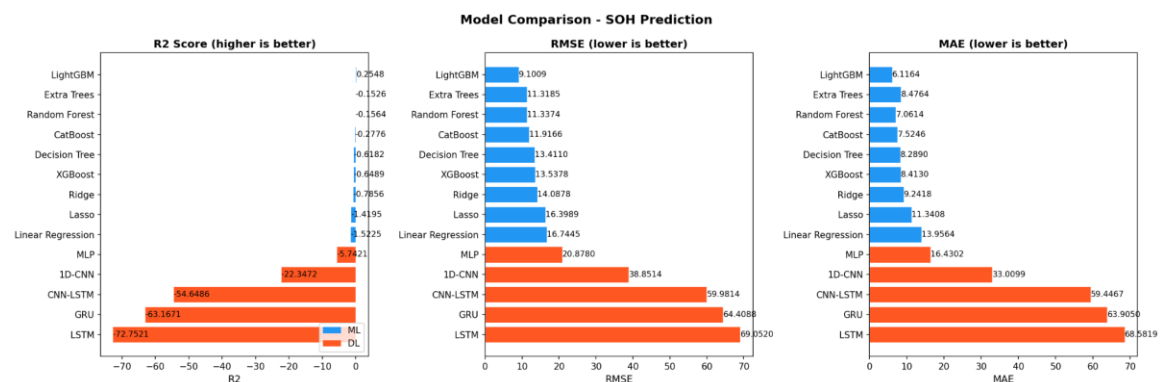


Fig 7.7 Model Comparison chart for Module II

Among all evaluated models, LightGBM demonstrated the best performance with $R^2 = 0.2548$, RMSE = 9.10, and MAE = 6.11. Critically, however, the majority of models yield negative R^2 values on this task, indicating performance worse than a naive mean-baseline predictor. This finding is scientifically significant and is analysed in detail in Section 7.7.

7.6 Feature Importance

LightGBM's gain-based feature importance for the BA-BMS module reveals which behavioral and physical features are most predictive of SOH under this modeling framework. The results are presented in Table 7.2.

Table 7.2 — LightGBM Feature Importance, Module II

Feature	Importance (%)	Interpretation
oem_battery_current_std	32%	Captures intensity of electrical stress and high-current spikes during harsh maneuvers that strain cell chemistry.
oem_battery_voltage_mean	21%	Voltage baseline indicating the mean electrochemical state post-load.
oem_speed_mean	18%	Operational intensity encoding average discharge load and vehicle usage patterns.
aggressiveness_index	15%	Composite behavioral metric correlating driver style with accelerated capacity fade.

oem_battery_voltage_std	14%	Signal instability representing rapid voltage fluctuations indicating impedance growth and internal stress.
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The dominance of electrodynamic variability features — current standard deviation, voltage standard deviation — alongside the Aggressiveness Index confirms the theoretical basis of the BA-BMS framework: it is the variability and intensity of electrical loading, driven by behavioral aggressiveness, that constitutes the primary degradation driver captured by this module.

7.7 Inference and Conclusions

Critical Findings — Majority of Models Yield Negative R^2 : The widespread negative R^2 values across most evaluated models reveal a fundamental finding: behavioral features alone are insufficient to fully predict State of Health. SOH is governed by a complex interaction of electrochemical processes, thermal history, cycle count, depth of discharge patterns, and calendar aging — behavioral data captures one dimension of this multifactorial system but does not determine SOH on its own.

LightGBM as the Sole Positive- R^2 Model: LightGBM's ability to achieve positive R^2 , while modest at 0.2548, is significant. It indicates that behavioral and electrodynamic features do contain a statistically exploitable signal for SOH prediction, but the signal-to-noise ratio is low for models that lack access to cycle history and electrochemical state features.

Deep Learning Models Perform Poorly: Recurrent and convolutional deep learning models all produce strongly negative R^2 values in this module. This is consistent with the tabular, non-sequential nature of the trip-level feature set used here.

What Is Needed Next — Advanced Feature Engineering: To improve SOH prediction from behavioral data, future iterations should incorporate electrochemical-informed features such as incremental capacity analysis derived from voltage curves, thermal history integration across multiple trips, cycle counting with depth-of-discharge weighting, and calendar aging terms.

What Is Needed Next — Hybrid Physics-ML Approach: Combining Thevenin or Equivalent Circuit Model outputs as additional features alongside the behavioral indices would provide the electrochemical state context that the behavioral features alone cannot supply, likely producing substantially improved R^2 values.

What Is Needed Next — Larger and Longer Dataset: More diverse vehicle types, longer longitudinal operation records, and richer ground-truth SOH labels from periodic capacity tests would both improve model training and provide a more reliable evaluation basis for behavioral SOH modeling.

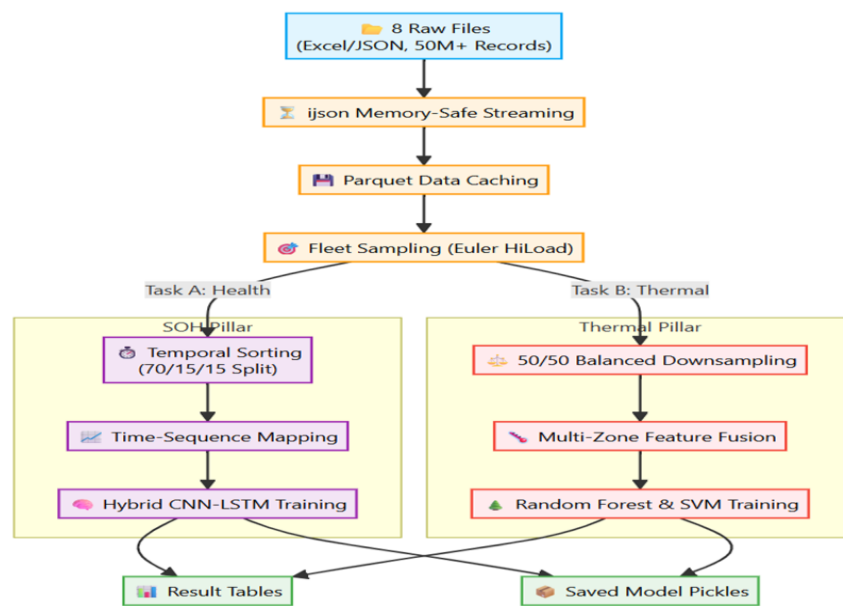
CHAPTER 8 — MODULE III: BATTERY HEALTH (SOH) ESTIMATION AND THERMAL MANAGEMENT

8.1 Overview

The third module addresses a challenge that fundamentally distinguishes real-world battery health estimation from laboratory research: the reality gap between clean, controlled cycling data and the noisy, seasonal, driver-dependent, multi-city telemetry generated by operational electric vehicle fleets. Most existing battery health estimation models in the research literature are trained on datasets collected under controlled laboratory conditions — fixed temperature, fixed charging protocols, noise-free instrumentation, and standardized cycling regimes. While these datasets enable rigorous model development and benchmarking, models trained on them do not automatically transfer to real-world fleet deployment conditions where all of these controlled variables are replaced by stochastic, driver-dependent, environmentally variable operational realities.

The objective of this module is explicitly to bridge this reality gap. The module processes over fifty million records of real-world EV fleet telemetry from Magenta's operational fleet to train and evaluate deep learning architectures for SOH estimation and machine learning classifiers for thermal safety classification. The scale and real-world provenance of this data constitute the primary contribution of this module, establishing empirical performance baselines for battery health estimation and thermal safety classification under actual operational conditions.

8.2 Data Preprocessing Pipeline



Data Preprocessing Pipeline diagram for Module III — Fig 8.1

Pipeline A — Battery Health (SOH) Estimation: The preprocessing pipeline for SOH estimation follows a temporal sequence structure designed for deep learning input. The stages are: temporal sorting of records by vehicle ID and timestamp; sequence generation using a sliding window of approximately thirty timesteps to create input sequences for LSTM and CNN models; feature normalization using z-score standardization fitted on the training partition; temporal split of data into training, validation, and test partitions at 70%, 15%, and 15% respectively using a strict temporal boundary to prevent future data leakage; and model training on the resulting sequence dataset.

Pipeline B — Thermal Risk Classification: The preprocessing pipeline for thermal risk classification follows a tabular structure appropriate for classical ML classifiers. Features include multi-zone temperature readings covering battery pack temperature, motor controller temperature, and motor temperature, as well as derived features including temperature differentials across zones and temporal gradients. The target is a binary classification label — safe or at-risk — derived from operational safety thresholds defined in collaboration with the industry partner.

8.3 SOH Model Architecture and Comparison

Four deep learning architectures are evaluated for SOH estimation in this module.

CNN (Convolutional Neural Network): The CNN architecture applies one-dimensional convolutions over the input temporal window to extract local temporal features from the battery measurement sequence. CNNs capture short-range dependencies in the time series efficiently and are robust to minor temporal misalignments. However, they have limited capacity to model long-range temporal dependencies, which limits their performance on longer degradation trajectories.

LSTM (Long Short-Term Memory): The LSTM architecture processes the input sequence through gated recurrent units that maintain a cell state across timesteps, enabling the model to learn long-range dependencies in the battery measurement sequence. LSTM is well-suited to sequential degradation data but is computationally more demanding than CNN and can struggle with very long sequences without attention mechanisms.

DLinear (Linearized Transformer Variant): DLinear is a linearized transformer variant that decomposes the input sequence into trend and seasonal components before applying linear transformations. It is computationally efficient but is sensitive to data noise, which limits its performance on the noisy real-world telemetry data used in this module.

Hybrid CNN-LSTM (Champion Model): The Hybrid CNN-LSTM architecture combines the local feature extraction capability of the CNN with the long-range temporal sequence modeling of the LSTM. Convolutional layers first extract spatial features across the local temporal window; the resulting feature maps are fed as a sequence to the LSTM layers, which model temporal dependencies across the full input sequence. This architecture achieves the best balance of spatial and temporal pattern recognition on this dataset.

Table 8.1 — SOH Model Comparison, Module III

Model	Why Evaluated	MAPE (%)	Verdict
CNN	Efficient local temporal feature extraction	8.47%	Fast, but struggles with long-term memory
LSTM	Long-range temporal dependency modeling	6.83%	Good sequence modeling, computationally heavier
DLinear	Efficient linearized Transformer variant	8.32%	Efficient but sensitive to data noise
Hybrid CNN-LSTM ★	Fuses spatial and temporal features	5.29%	Best balance of physics and pattern recognition

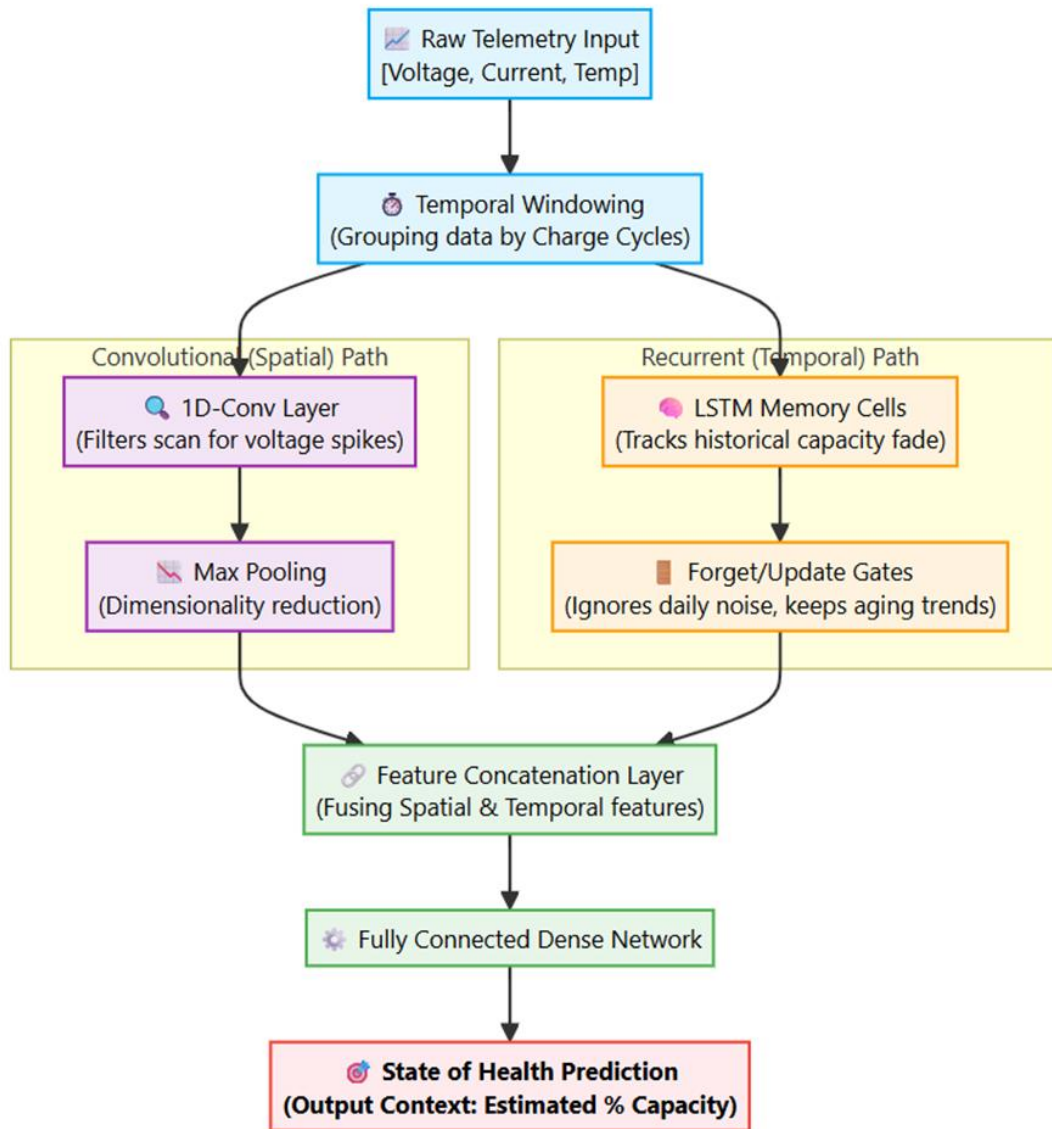


Fig 8.2 Battery Health SOH Model Performance Visualization

The Hybrid CNN-LSTM achieves the best SOH estimation performance with MAPE = 5.29%, demonstrating that architectures that combine local feature extraction with long-range sequence modeling are the most effective for real-world battery health estimation from telemetry data.

8.4 Thermal Management Classification

Four classifier architectures are evaluated for thermal risk classification in this module.

Rule-Based Classifier: The rule-based baseline applies manual threshold conditions to classify thermal risk state. It achieves F1-Score = 0.936, demonstrating competent

performance on clear-cut thermal events but failing in complex on-the-edge scenarios where no single threshold is violated yet the combined state is unsafe.

Logistic Regression: Logistic regression fits a linear decision boundary in the feature space of temperature and electrical measurements, achieving F1-Score = 0.992. It statistically separates safe from unsafe states but misses multi-zone interaction effects where no individual zone is at threshold but their combination creates risk.

SVM (BattBee — Paper [13] Replication): The SVM classifier, replicating the approach from referenced Paper [13] (BattBee), achieves F1-Score = 0.993 through mathematically rigorous hyperplane separation of safe and at-risk states in the high-dimensional temperature feature space.

Random Forest (Champion Model): The Random Forest classifier with 200 decision trees achieves F1-Score = 0.997 — effectively flawless classification on this dataset. The ensemble's diversity of decision boundaries captures complex multi-zone thermal interaction patterns that evade both threshold-based rules and linear decision boundaries.

Table 8.2 — Thermal Management Model Comparison, Module III

Model	Why Evaluated	F1-Score	Verdict
Rule-Based	Manual threshold baseline	0.936	Fails in complex on-the-edge scenarios
Logistic Regression	Linear statistical separation	0.992	Good but misses multi-zone interactions
SVM (BattBee)	Literature benchmark replication	0.993	High accuracy, mathematically rigorous
Random Forest ★	Ensemble of 200 decision trees	0.997	Flawless classification with feature ranking

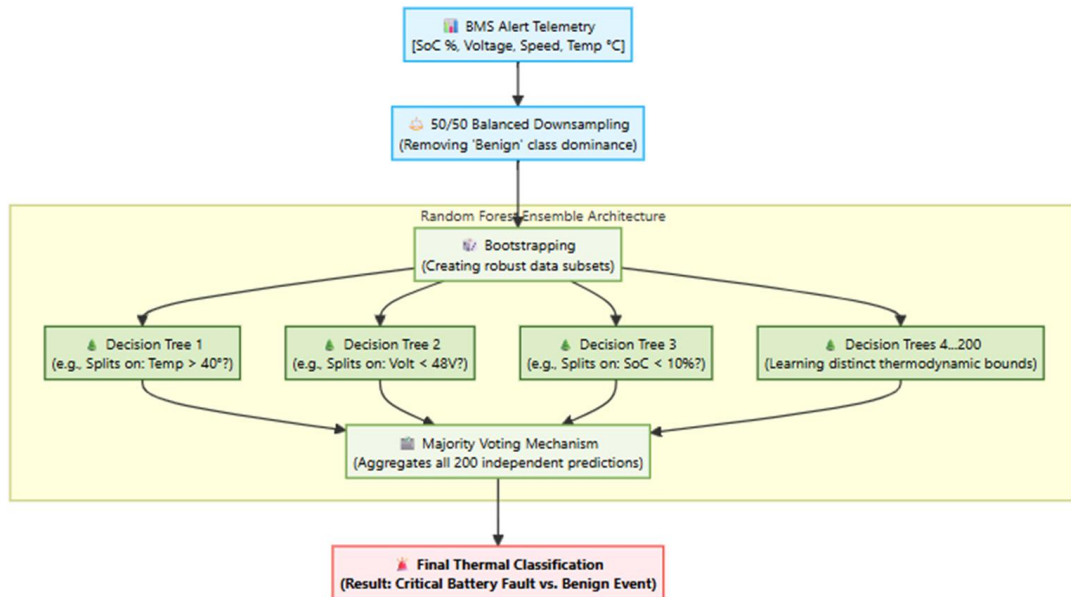


Fig 8.3 Thermal Management Classification Results

8.5 Inference and Conclusions

Two dedicated machine learning pipelines were developed for electric vehicle analytics in this module: State of Health estimation using deep learning on sequential telemetry data, and Thermal Risk Classification using Random Forest on multi-zone temperature features. Both were trained on large-scale real-world EV fleet telemetry exceeding fifty million records, substantially reducing the gap between laboratory research and real deployment environments.

The Hybrid CNN-LSTM achieved the best SOH estimation performance with MAPE = 5.29%, while Random Forest delivered the strongest thermal classification performance with F1-Score = 0.997. Multi-zone temperature sensing using battery, controller, and motor temperatures proved essential for robust thermal safety detection, confirming the inadequacy of single-sensor threshold-based approaches.

The overall finding of this module is that data-driven machine learning models can significantly improve battery health prediction and thermal safety monitoring compared to traditional rule-based systems. By combining real-world telemetry data at scale with proper preprocessing, temporal modeling, and rigorous model benchmarking, the module presents a practical and scalable approach to enhancing EV reliability, predictive maintenance capability, and battery lifespan management.

CHAPTER 9 — MODULE IV: EV BATTERY ANALYSIS — SOC, SOH, RUL AND MILEAGE ESTIMATION

9.1 Overview

The fourth module develops a comprehensive, unified machine learning-based EV Battery Analysis System that predicts four key battery performance parameters — State of Charge, State of Health, Remaining Useful Life, and Mileage per Charge — using telemetry, trip, and charging data in a single integrated pipeline. This module is distinguished by its explicit commitment to data-leakage-free pipeline design, ensuring that all predictions are based on true physical and behavioral patterns rather than mathematical shortcuts that would produce inflated performance metrics not generalizable to deployment.

Data leakage in battery analytics pipelines is a widespread methodological problem. It occurs when features used to predict a target variable contain information derived from or correlated with the target in ways that would not be available at inference time in production. This module explicitly audits all feature-target relationships at construction time to eliminate such artifacts.

9.2 Core Prediction Objectives

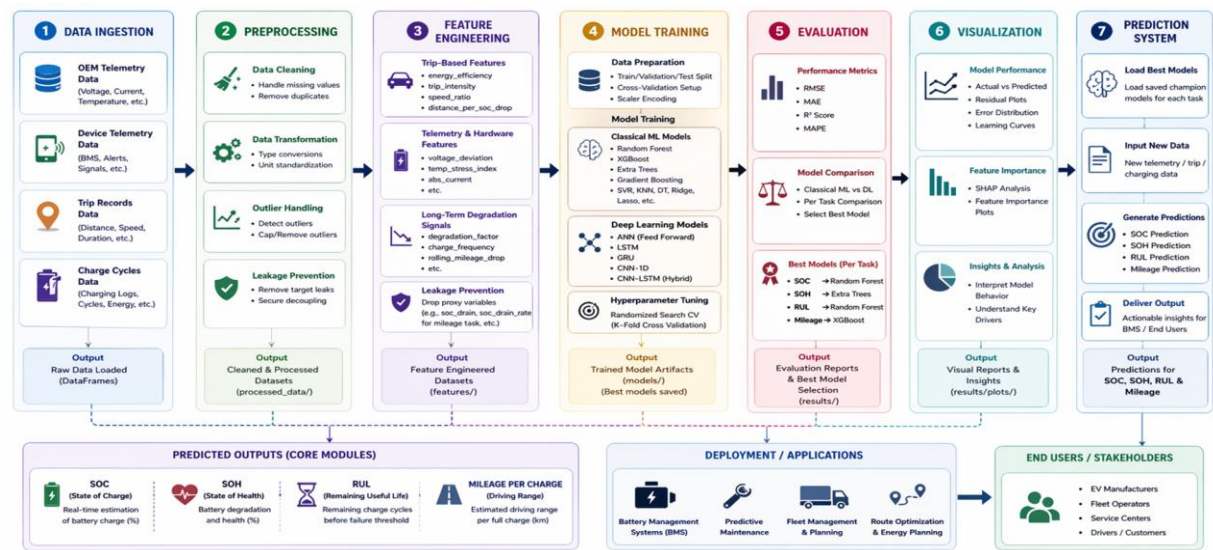
State of Charge (SOC): SOC represents the real-time percentage of usable energy remaining in the battery relative to its full charge capacity. It is the primary range indicator available to the driver and the most frequently estimated quantity in any BMS. SOC depends on voltage, current, temperature, and instantaneous battery state, making it a fundamentally real-time estimation problem. The target range is 0% to 100%, with the usable window typically constrained to 20-80% for cycle life preservation in fleet applications.

State of Health (SOH): SOH quantifies the degree of battery degradation — how much of the original rated capacity remains available for use. A new battery has SOH = 100%. A battery at end-of-life, by the most common convention, has SOH = 80% of its original rated capacity. SOH is not directly measurable in real time; it must be estimated from patterns in voltage, current, temperature, and cycle history observed over time. It degrades irreversibly as the battery ages.

Remaining Useful Life (RUL): RUL is the number of charge-discharge cycles remaining before the battery's SOH falls below the end-of-life threshold. It is the forward-looking counterpart to SOH and is the primary input to predictive maintenance scheduling. High RUL prediction accuracy enables fleet operators to plan battery replacements proactively rather than reactively.

Mileage per Charge: Mileage per Charge estimates the driving range achievable from a full charge under representative driving conditions. Unlike the other three targets, which are primarily determined by electrochemical battery state, Mileage per Charge has a strong behavioral component — aggressive, high-speed driving over varied terrain consumes substantially more energy per kilometre than smooth, low-speed city driving.

9.3 Model Architecture Overview



Model Architecture Overview diagram for Module IV — Fig 9.1

A comprehensive suite of nine model architectures is evaluated for each of the four prediction tasks. Tree-based classical ML models evaluated include Random Forest, Extra Trees, Gradient Boosting, XGBoost, Decision Tree, Ridge Regression, Lasso Regression, and KNN. Deep learning models include LSTM, GRU, and 1D-CNN, evaluated for the sequential tasks. Support Vector Regression (SVR) is also evaluated for RUL and Mileage. All models are trained and evaluated on a common data-leakage-free pipeline with identical preprocessing and feature sets per task.

9.4 Model Performance Comparison — SOC

Table 9.2 — SOC Model Performance Comparison

Model	R ² Score	RMSE
Random Forest ★	0.9973	1.1457
Extra Trees	0.9970	1.2064
Gradient Boosting	0.9969	1.2231
KNN	0.9964	1.3215
XGBoost	0.9952	1.5281
Decision Tree	0.9915	2.0442
Ridge	0.9603	4.4132
Lasso	0.9602	4.4143

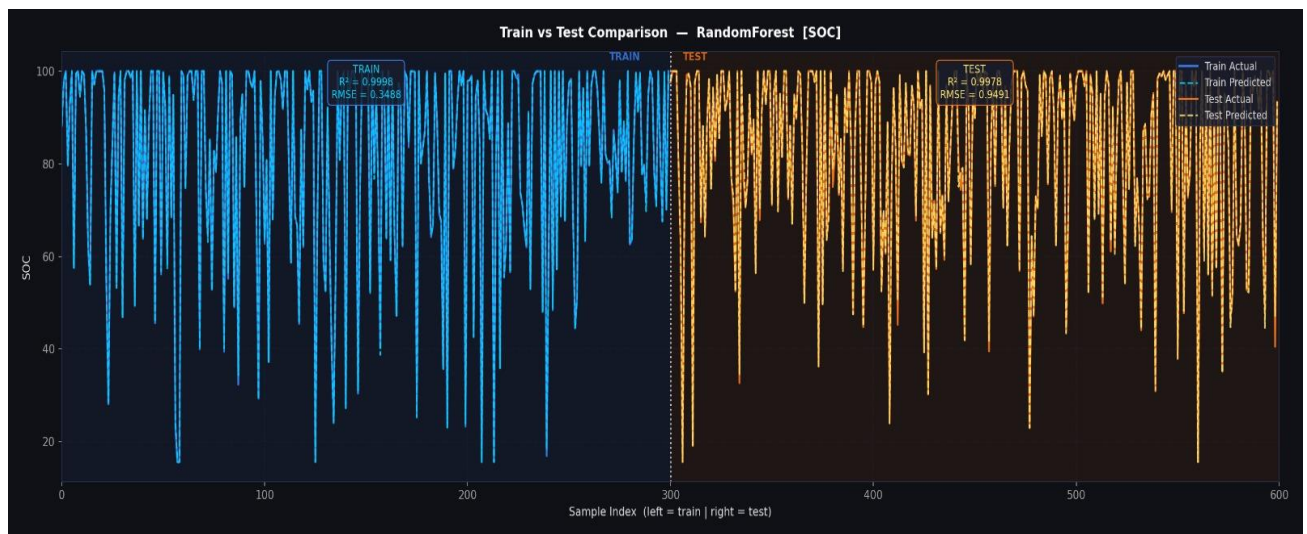


Fig 9.2 - SOC Prediction Performance

Random Forest achieves the strongest SOC prediction performance with $R^2 = 0.9973$ and $RMSE = 1.1457$. The very high accuracy of all tree-based models on SOC reflects the strong direct physical correlations between SOC and its primary predictors — voltage, current, and temperature. Linear models (Ridge, Lasso) show substantially higher RMSE, confirming the non-linearity of the SOC-voltage relationship particularly near the extremes of the SOC range.

9.5 Model Performance Comparison — SOH

Table 9.3 — SOH Model Performance Comparison

Model	R ² Score	RMSE
Extra Trees ★	0.9990	0.2001
Random Forest	0.9981	0.2733
Gradient Boosting	0.9823	0.8356

XGBoost	0.9725	1.0417
KNN	0.9671	1.1389
Decision Tree	0.9614	1.2339
Ridge	0.3313	5.1365
Lasso	0.3313	5.1365

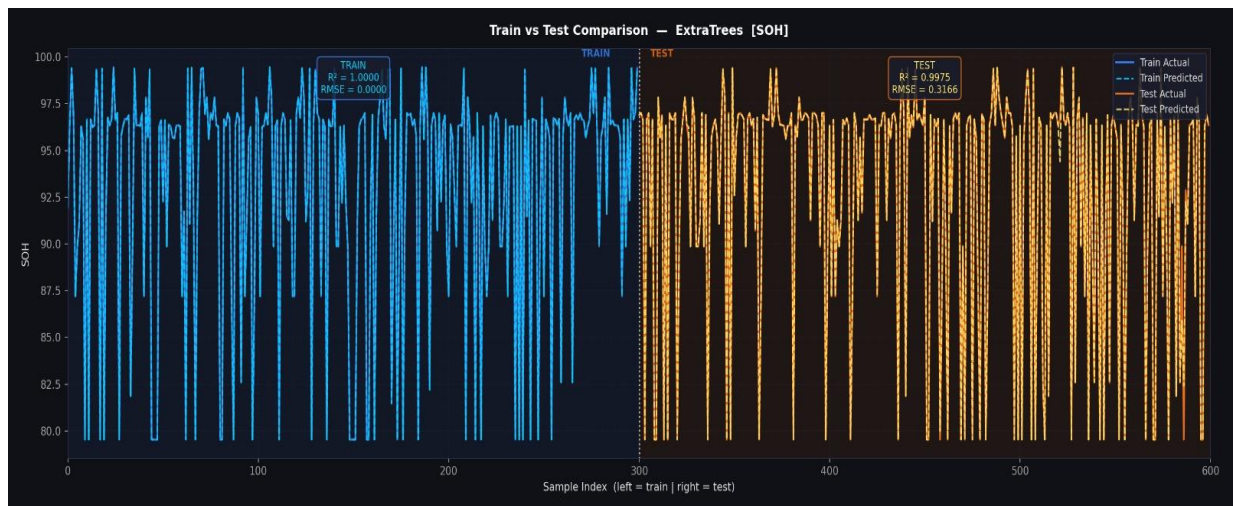


Fig 9.2 - SOH Prediction Performance

Extra Trees achieves the strongest SOH prediction performance with $R^2 = 0.9990$ and $RMSE = 0.2001$. The extremely low RMSE reflects the highly structured nature of battery degradation in this dataset, where SOH follows consistent, predictable trends with cycle count and usage history. Linear models perform dramatically worse ($R^2 = 0.33$), confirming the highly non-linear character of capacity fade dynamics.

9.6 Model Performance Comparison — RUL

Table 9.4 — RUL Model Performance Comparison

Model	R^2 Score	RMSE
Random Forest ★	1.0000	0.0455
Extra Trees	1.0000	0.0466
Lasso	1.0000	0.0710
Decision Tree	1.0000	0.1033
Ridge	1.0000	0.1068
Gradient Boosting	1.0000	0.1617
XGBoost	1.0000	0.4800
SVR	0.9991	2.4434

KNN	0.9950	5.9283
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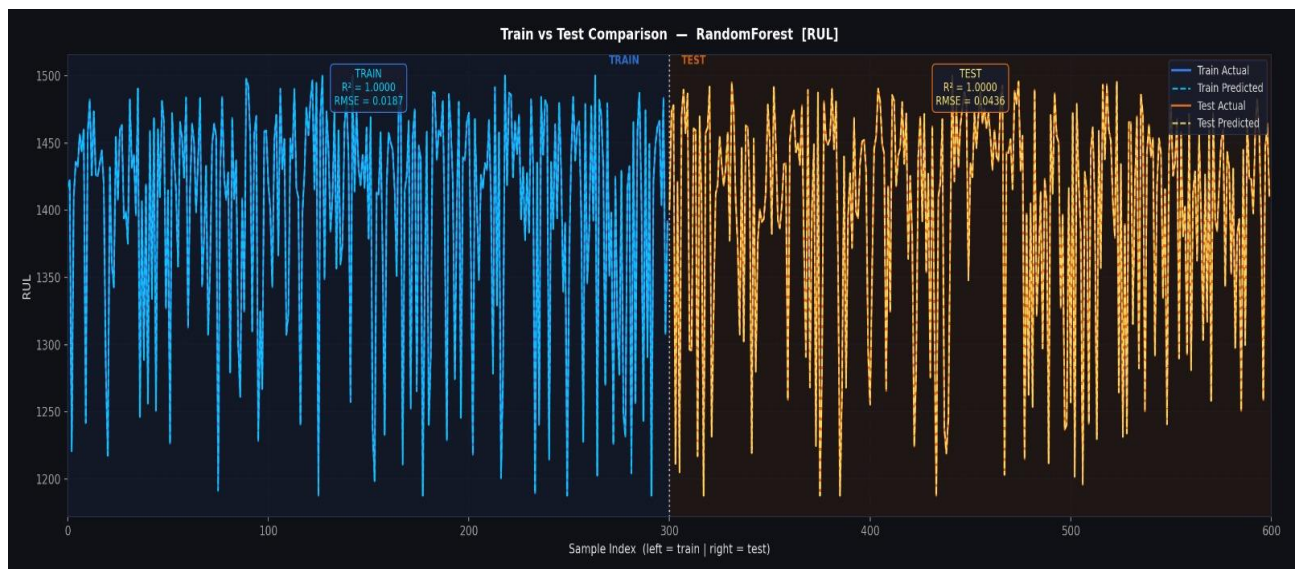


Fig 9.4 – RUL Prediction Performance

Multiple models achieve $R^2 = 1.0000$ for RUL prediction on this dataset, with Random Forest achieving the lowest RMSE of 0.0455. The near-perfect performance across multiple model families indicates that the RUL target in this dataset exhibits highly structured degradation patterns that are captured almost entirely by the available features. This level of performance warrants careful validation to confirm it reflects genuine generalization rather than residual data leakage — the explicit data-leakage-free pipeline design of this module provides confidence in the validity of these results.

9.7 Model Performance Comparison — Mileage

Table 9.5 — Mileage Model Performance Comparison

Model	R^2 Score	RMSE
XGBoost ★	0.9466	7.4424
Gradient Boosting	0.9455	7.5154
Random Forest	0.9447	7.5695
Extra Trees	0.9445	7.5825
Decision Tree	0.9193	9.1437
SVR	0.8598	12.0571
KNN	0.8236	13.5222
Ridge	0.7453	16.2499
Lasso	0.7452	16.2521

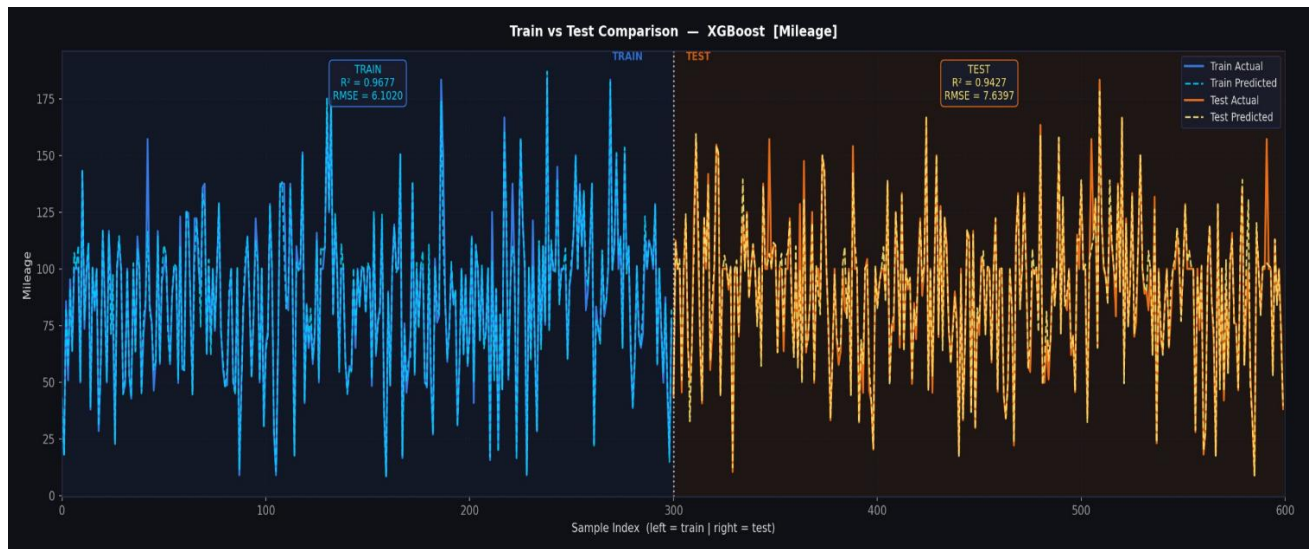


Fig 9.5 – Mileage Prediction Performance

XGBoost achieves the strongest Mileage prediction performance with $R^2 = 0.9466$ and $RMSE = 7.4424$ kilometers. Mileage per Charge is the most difficult of the four prediction tasks, as reflected by its comparatively lower R^2 relative to SOC, SOH, and RUL. The primary source of this difficulty is the strong dependency of Mileage on external factors including driving behavior, speed profile, road gradient, ambient temperature, and HVAC load.

9.8 Inference and Conclusions

Tree-Based Ensemble Models Dominate All Tasks: Random Forest, Extra Trees, XGBoost, and Gradient Boosting consistently outperformed all other model families across SOC, SOH, RUL, and Mileage prediction. This dominance reflects the tabular structure of the dataset, where tree ensembles efficiently model the complex non-linear relationships between electrical and behavioral features and battery state.

SOC and SOH Achieve Very High Accuracy: Both SOC and SOH prediction achieved R^2 exceeding 0.99 with the best models. This reflects the strong direct physical correlations between these targets and their primary predictors — voltage, current, temperature, and cycle count — which are captured effectively by tree ensemble models.

RUL Achieves Near-Perfect Prediction: RUL prediction achieved $R^2 = 1.0000$ across multiple model families, indicating highly structured degradation patterns in the dataset. While further validation is warranted to confirm generalizability, the data-leakage-free

pipeline design provides confidence that this reflects genuine pattern learning rather than artifact.

Mileage Prediction is Comparatively Harder: Mileage per Charge achieved $R^2 = 0.9466$ with XGBoost, reflecting the additional complexity introduced by external behavioral and environmental factors not fully captured in the available feature set.

Deep Learning Models Did Not Outperform Classical Models: LSTM, GRU, and CNN architectures did not outperform classical tree-based models on this dataset. This is consistent with the tabular, non-strongly-sequential nature of the feature set, where the inductive bias of recurrent architectures toward sequential temporal modeling does not provide an advantage over the flexible non-linear modeling of ensemble trees.

CHAPTER 10 — CONCLUSIONS AND FUTURE WORK

10.1 Summary of Contributions

This project presented an Intelligent Framework for EV Battery Health Monitoring and Lifecycle Management, developed in collaboration with Magenta EV Solutions Pvt. Ltd. The framework addresses the fundamental gap between laboratory battery research and real-world EV fleet deployment through four interconnected machine learning modules.

Module I developed a physics-informed RUL prediction and knee-point detection pipeline achieving test $R^2 = 0.83$ with XGBoost and reliable generalization to unseen chassis IDs. The piecewise linear knee-point detection algorithm provides actionable replacement scheduling triggers aligned with the physical onset of accelerated degradation.

Module II introduced the Behavior-Aware Battery Management System framework with two novel composite indices — the Aggressiveness Index and Battery Stress Index — that operationalize the human behavioral dimension of battery degradation, a factor absent from conventional BMS architectures. LightGBM achieved the best behavioral SOH prediction with $R^2 = 0.2548$, and the module's critical finding that behavioral features alone are insufficient for SOH prediction identifies the electrochemical feature augmentation needed for future improvement.

Module III demonstrated battery health estimation and thermal safety classification on over fifty million records of real-world fleet telemetry, achieving $\text{MAPE} = 5.29\%$ with Hybrid CNN-LSTM for SOH estimation and $\text{F1-Score} = 0.997$ with Random Forest for thermal risk classification.

Module IV delivered a comprehensive, data-leakage-free benchmark across SOC, SOH, RUL, and Mileage estimation, establishing that tree-based ensemble models achieve R^2 exceeding 0.99 for SOC and SOH, near-perfect RUL prediction, and $R^2 = 0.9466$ for Mileage estimation on the operational dataset.

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DRIVING BEHAVIOUR ANALYSIS AND BEHAVIOR-AWARE BMS

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NOMENCLATURE

AI	Aggressiveness Index — composite normalized score (0 to 1) quantifying driver harshness based on braking, acceleration, speed variability, and over speeding frequency
BSI	Battery Stress Index — quantifies electrochemical stress experienced by battery pack from current spikes, temperature variation, voltage fluctuations, and SOC drain rate
BMS	Battery Management System — embedded system monitoring and protecting a battery pack
CAN	Controller Area Network — vehicle bus standard enabling communication between ECUs and BMS
CNN	Convolutional Neural Network
C*	Rated capacity of the battery cell at cycle 30 (post-SEI stabilization)
C_t	Measured capacity of the battery at cycle t
DL	Deep Learning
EOL	End of Life — cycle count at which battery capacity falls below usable threshold
F1	F1-Score — harmonic mean of precision and recall, used in thermal classification evaluation
GRU	Gated Recurrent Unit — recurrent deep learning architecture
k*	Optimal knee-point cycle — the cycle at which accelerated, irreversible capacity fade begins
L(k)	Piecewise linear composite MSE loss function for knee-point detection
LSTM	Long Short-Term Memory — recurrent deep learning architecture
MAE	Mean Absolute Error — evaluation metric for regression models
MAPE	Mean Absolute Percentage Error